

Semester Thesis in Mechatronics, Robotics, and Biomechanical Engineering

Hydrodynamic Modeling and Trajectory Optimization of Paddling Propulsion for a Quadruped Robot

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Abstract

Amphibious quadruped robots can swim by moving their legs, but a leg designed for walking makes a poor propulsor. This thesis asks which propulsion principle suits the legs of the quadruped BODY2 and how each propulsor should move. Drag-based paddling with a folding fan paddle is compared with lift-based flapping of a small fluke fitted to the same leg. Three simulation tools with different roles are used. First, a reduced-order MuJoCo model with Morison-type link forces and an actuator speed limit is combined with CMA-ES gait optimization. Whenever the model contains lift and lift is not suppressed, every feasible optimized gait is lift-propelled, and at equal speed the lift gaits need one half to one third of the power of the best gaits of a drag-only model. Second, the single leg is simulated with overset-grid CFD in OpenFOAM. For the fan paddle, a trapezoidal velocity law, a 12-mm folded width and closing the fan at the start of the deceleration raise the thrust from 23.4 mN for the robot's current gait to 73–83 mN per leg at rest; the range bounds the change of added mass at the fold, which the two-state model of the folding fan does not resolve. The paddle loses its thrust by about 0.30 m s^{-1} , because its recovery drag and the cost of opening the fan in the oncoming flow grow while its power-stroke thrust decays. The fluke produces the same thrust per area at rest and loses it more slowly with speed. Thrust and efficiency cross over at about 0.24 and 0.22 m s^{-1} . Scheduling the fluke pitch with speed keeps the angle of attack near 20° and raises the Froude efficiency at 0.30 m s^{-1} from 0.24 to 0.40. A quasi-steady self-propulsion balance predicts a cruising speed of 0.30 m s^{-1} for the fluke and $0.26\text{--}0.27 \text{ m s}^{-1}$ for the optimized paddle, with 43 % less power for the fluke with speed-scheduled pitch. A first hardware test with flukes at 2 Hz is 1.3 to 1.6 times slower than predicted, so these speeds are upper bounds for the present robot. Third, the real-time solver Flare Live supplies the reference fluke motion, but its lattice does not resolve the foil and it underpredicts the fluke thrust, so it is used for trends only. The CFD set-up reproduces the thrust coefficient of a flapping-foil experiment to within 8 %.

Kurzfassung

Amphibische Vierbeiner können schwimmen, indem sie ihre Beine bewegen, doch ein zum Laufen ausgelegtes Bein ist ein schlechter Antrieb. Diese Arbeit untersucht, welches Antriebssprinzip zu den Beinen des Vierbeiners BODY2 passt und wie sich der jeweilige Antrieb bewegen sollte. Verglichen werden widerstandsbasiertes Paddeln mit einem faltbaren Fächerpaddel und auftriebsbasiertes Schlagen mit einer kleinen Flosse am selben Bein. Dazu werden drei Simulationswerkzeuge mit unterschiedlichen Aufgaben eingesetzt. Zunächst wird ein Modell reduzierter Ordnung in MuJoCo mit Morison-Kräften je Glied und begrenzter Stellgeschwindigkeit der Servos mit einer CMA-ES-Gangoptimierung kombiniert. Sobald das Modell Auftrieb enthält und dieser nicht unterdrückt wird, nutzen alle zulässigen optimierten Gangarten Auftrieb, und bei gleicher Geschwindigkeit benötigen sie nur die Hälfte bis ein Drittel der Leistung

der besten Gangarten eines reinen Widerstandsmodells. Anschließend wird das Einzelbein mit Overset-CFD in OpenFOAM simuliert. Für das Fächerpaddel erhöhen ein trapezförmiges Geschwindigkeitsprofil, eine gefaltete Breite von 12 mm und das Schließen des Fächers zu Beginn der Verzögerung den Schub im Stand von 23,4 mN (aktueller Gang des Roboters) auf 73–83 mN pro Bein; die Spanne erfasst die Änderung der hydrodynamischen Masse beim Falten, die das Zwei-Zustands-Modell des Fächers nicht auflöst. Bei etwa $0,30 \text{ m s}^{-1}$ verliert das Paddel seinen Schub, weil Rückholwiderstand und der Aufwand zum Öffnen des Fächers in der Anströmung wachsen, während der Schub im Arbeitshub abnimmt. Die Flosse erzeugt im Stand denselben Schub pro Fläche und verliert ihn mit der Geschwindigkeit langsamer. Schub und Wirkungsgrad kreuzen sich bei etwa $0,24$ bzw. $0,22 \text{ m s}^{-1}$. Eine geschwindigkeitsabhängige Anstellung hält den Anstellwinkel bei etwa 20° und steigert den Froude-Wirkungsgrad bei $0,30 \text{ m s}^{-1}$ von $0,24$ auf $0,40$. Eine quasistationäre Eigenantriebsbilanz sagt für die Flosse $0,30 \text{ m s}^{-1}$ und für das optimierte Paddel $0,26$ – $0,27 \text{ m s}^{-1}$ voraus, wobei die Flosse mit geschwindigkeitsabhängiger Anstellung 43 % weniger Leistung benötigt. Ein erster Schwimmversuch des Roboters mit Flossen ist bei 2 Hz 1,3- bis 1,6-mal langsamer als vorhergesagt; die vorhergesagten Geschwindigkeiten sind daher Obergrenzen für den aktuellen Roboter. Der Echtzeitleser Flare Live liefert die Referenzbewegung der Flosse, löst das Profil jedoch nicht auf und unterschätzt den Flossenschub; er dient daher nur zur Einordnung von Trends. Der CFD-Aufbau gibt den Schubbeiwert eines Schlagflügelexperiments auf 8 % genau wieder.

Contents

List of Figures	viii
List of Tables	ix
List of Acronyms and Symbols	x
1 Introduction	1
1.1 Motivation	1
1.2 Problem Statement and Scope	2
1.3 Research Questions and Hypotheses	3
1.4 Contributions	4
1.5 Thesis Outline	4
2 Related Work	5
2.1 Drag- and Lift-Based Swimming in Animals	5
2.2 Flapping-Foil Hydrodynamics	6
2.3 Paddling Mechanics	6
2.4 Swimming Legged Robots	6
2.5 Simulation Approaches for Swimming Robots	7
2.6 Research Gap	8
3 Fundamentals	9
3.1 Forces on a Moving Appendage	9
3.2 Quasi-Steady Paddling	10
3.3 Flapping-Foil Kinematics	10
3.4 Performance Metrics	11
3.5 Self-Propulsion Balance	12
4 Robot Platform and Propulsors	13
4.1 The Quadruped BODY2	13
4.2 Propulsors	13
4.3 Leg Motions	14
4.4 Hardware Prototype	16

5	Reduced-Order Gait Optimization	18
5.1	Model	18
5.1.1	Robot	18
5.1.2	Hydrodynamic Forces	18
5.1.3	Servo Model	20
5.2	Gait Parametrization	20
5.3	Objective, Constraints and Optimizer	21
5.4	Results	22
5.4.1	Baseline and Servo Limit	22
5.4.2	Drag versus Lift at Equal Speed	22
5.4.3	Drag-Based Strokes within the Lift Model	22
5.5	Summary	23
6	Flow Simulation Methods	25
6.1	Single-Leg CFD with Overset Grids	25
6.1.1	Governing Equations and Solver	25
6.1.2	Domain, Boundary Conditions and Motion	25
6.1.3	Meshes	26
6.1.4	Two-State Composite for the Folding Fan	26
6.1.5	Post-Processing	27
6.1.6	Case Matrix	28
6.2	Whole-Robot Simulation in Flare Live	28
7	Verification and Validation	30
7.1	Validation with a Heaving and Pitching Foil	30
7.2	Mesh Sensitivity	31
7.3	Temporal Convergence	31
7.4	Consequences for the Drag-Type Cases	31
7.5	Cross-Check of the Real-Time Solver	32
7.6	Summary	33
8	Results	34
8.1	Design of the Drag-Type Stroke	34
8.2	The Lift-Type Fluke with a Fixed Pitch Schedule	35
8.3	Drag versus Lift across Speed	37
8.4	Speed-Scheduled Fluke Pitch	38
8.5	Self-Propulsion Estimate	39
8.6	First Hardware Test	41
9	Discussion	42
9.1	Evaluation of the Hypotheses	42
9.2	Why the Reduced-Order Model and the CFD Differ at Low Speed	43
9.3	Implications for BODY2	44
9.4	Limitations	45
10	Conclusion and Outlook	46
10.1	Conclusion	46
10.2	Outlook	47

A	Additional Data	48
A.1	Drag-Type Paddle: Power Stroke and Recovery	48
A.2	Fluke Cases	48
A.3	Actuator Requirements	49
B	Inter-Limb Coordination in the Reduced-Order Model	50
B.1	Gait Families without Lift	50
B.2	Coordination with Lift under Strict Limits	50
B.3	Electrical Power	51
C	Real-Time Solver: Tow Runs and Whole-Robot Trends	54
C.1	Tow Runs	54
C.2	Self-Propelled Runs	54
D	Reproducibility	56
D.1	Software	56
D.2	MuJoCo Settings	57
D.3	Model Errors Found and Corrected	57
	Bibliography	58

List of Figures

1.1	Overview of the method	3
3.1	Flapping-foil angles	11
4.1	BODY2 with fan paddles and with flukes	14
4.2	Propulsor geometries	14
4.3	Leg motions	16
4.4	BODY2 hardware with flukes	17
5.1	MuJoCo model	19
5.2	Least power at equal speed	23
6.1	Two-state composite	28
7.1	Verification and validation	32
7.2	Flare Live compared with CFD	33
8.1	Wake of the fluke	36
8.2	Thrust, power and efficiency versus speed	37
8.3	Power stroke and recovery of the paddle	38
8.4	Speed-scheduled pitch	39
8.5	Self-propulsion estimate	40
B.1	Fastest gaits per family in MuJoCo	51
B.2	A lift-propelled trot in MuJoCo	52
B.3	Least power per gait family	53
C.1	Flare Live free-swimming trends	55

List of Tables

2.1	Swimming legged robots	7
4.1	Propulsor data	15
5.1	MuJoCo robot model	19
5.2	Least power at equal speed, drag versus lift	22
6.1	Mesh parameters	26
6.2	Single-leg simulation cases	28
7.1	Validation against Schouveiler et al.	31
7.2	Uncertainty summary	33
8.1	Design steps of the drag-type stroke	35
8.2	Fluke with the L0 motion	36
8.3	Propulsor comparison across speed	37
8.4	Speed-scheduled pitch	39
8.5	Predicted cruising speed	40
8.6	First hardware test	41
10.1	Key findings	46
A.1	Power-stroke and recovery contributions	48
A.2	All fluke cases	48
A.3	Actuator requirements	49
B.1	Fastest gait per family	51
C.1	Flare Live tow runs	54
C.2	Flare Live self-propelled runs	55
D.1	MuJoCo settings	57

List of Acronyms and Symbols

AR	aspect ratio, $AR = b^2/S$
BL	body length (speeds in BL/s)
CFD	computational fluid dynamics
CMA-ES	covariance matrix adaptation evolution strategy
COT	cost of transport
EMF	electromotive force (back-EMF of the servo motor)
GCI	grid convergence index
LSTM	long short-term memory (recurrent neural network)
MJCF	MuJoCo XML model format
PIMPLE	merged PISO–SIMPLE pressure–velocity coupling algorithm
RMS	root mean square
SPH	smoothed-particle hydrodynamics
URDF	unified robot description format

U	forward (swimming or tow) speed, m s^{-1}
f, T	stroke frequency and period, $T = 1/f$
τ	cycle phase $t/T \bmod 1$
$\bar{T}$	cycle-mean thrust of one leg, mN
$\bar{P}$	cycle-mean hydrodynamic input power of one leg, mW
η	Froude efficiency $\bar{T}U/\bar{P}$
C_T	thrust coefficient $\bar{T}/(\frac{1}{2}\rho U^2 S)$
St	Strouhal number fA/U with peak-to-peak excursion A
α	angle of attack between foil chord and relative inflow
γ	inflow angle of the heaving foil, $\gamma = \arctan(\dot{h}/U)$
θ	fluke pitch angle

DUTY	fraction of the cycle spent in the power stroke
τ_o, τ_c	phases at which the fan paddle is opened and closed
K	hull drag coefficient in $D = KU U $, $\text{Ns}^2 \text{m}^{-2}$
ρ, ν	density and kinematic viscosity of water

Case names

Robot gait	fan paddle with the robot's current stroke (1.25 s period, fold to 30 mm)
V1–V3	design steps of the drag-type stroke at rest; V3trap: the optimized stroke (trapezoidal velocity law, fold to 12 mm, early closing)
L0	fluke with the reference motion from Flare Live at 0.223 m s^{-1} ; L0c: the same on the coarse mesh
L2	L0 motion at other speeds (0 m s^{-1} , 0.08 m s^{-1} , 0.15 m s^{-1} and 0.30 m s^{-1})
L1	fluke with an angle-of-attack law (L1a10, L1a28: targets of 10 and 28°)
L3	fluke with the speed-scheduled pitch $\theta_0 = \gamma_{\max}/2$ (L3U030, L3U040 at 0.30 and 0.40 m s^{-1})
Validation	heaving and pitching NACA 0012 foil after Schouveiler et al. (Section 7.1)

Chapter 1

Introduction

1.1 Motivation

Legged robots move well on land, and many of the places where they are useful, such as shorelines, flooded areas, ponds and river banks, also contain open water. A quadruped that crosses such water needs some way to propel itself when its feet no longer touch the ground. Adding thrusters is one option [Yao+23], but thrusters add mass, volume and actuators that are useless on land. The alternative is to swim with the legs themselves [LFC+19; Qu+25; Han+25; Wan+25a]. Then no extra hardware is needed, and swimming becomes a question of how the existing joints are moved and what the feet look like.

Animals that swim with legs use two different physical principles [Fis96]. A dog, a muskrat or a duck *paddles*. It pushes water backwards with a foot oriented normal to its motion and then brings the foot forward again [Fis84; FDT21]. The thrust comes from drag on the foot during the power stroke, and the recovery stroke unavoidably costs some thrust. Dolphins, sea turtles and many fish instead *flap* a fin or fluke across the swimming direction. The fin meets the oncoming water at a small angle of attack and produces lift, whose forward component is the thrust. Drag-based paddling is simple, works at rest and accelerates well. Lift-based flapping is two to three times more efficient at cruising speed [Fis96; WW00]. Mammals that became fully aquatic changed over from the first principle to the second.

Most swimming legged robots use the first principle, because a walking foot is already a paddle. This thesis studies whether that choice is right for a small amphibious quadruped, BODY2, and how the leg should move for each principle. The robot's legs were designed for walking. Each is a planar two-degree-of-freedom linkage driven by two hobby servos, and the foot is a fan-shaped paddle that folds during the recovery stroke. Its current swimming gait was taken from video of the robot and has not been designed for thrust. Whether this paddle can be made efficient, or whether a small lift-producing fluke on the same leg would be better, cannot be decided from the literature alone, because the answer depends on the leg geometry, the actuator limits and the swimming speed.

1.2 Problem Statement and Scope

The problem is to find, for a given leg and given actuators, the propulsor and the periodic leg motion that propel the robot most effectively, and to quantify how the answer depends on swimming speed. Three properties make the problem difficult.

First, the flow is unsteady and separated. A paddle starts and stops twice per cycle, sheds a vortex at every stop, and meets its own wake. A flapping fluke works at Reynolds numbers of 10^3 – 10^4 , where leading-edge vortices and the phase between heave and pitch decide the thrust [TTG91; And+98]. Quasi-steady models capture trends but not the magnitudes of these effects [KG11].

Second, the answer depends on speed. At rest, a paddle pushes water that is not moving and produces thrust over most of its power stroke. At speed, the oncoming flow reduces the relative velocity during the power stroke and increases it during the recovery. A lift-based foil behaves in the opposite way: its angle of attack is set by the ratio of heave velocity to swimming speed, so it depends on speed through the kinematics. A comparison at a single speed can therefore favour either principle.

Third, no single model is both accurate and cheap. A three-dimensional CFD simulation of one leg takes hours to days and cannot be placed inside an optimizer that needs thousands of evaluations. A reduced-order model can, but its coefficients are not calibrated and its lift model is a textbook flat plate.

Three simulation tools are therefore used, each for a different question, and none is expected to reproduce the numbers of the others:

- a *reduced-order* whole-robot model in MuJoCo with CMA-ES gait optimization answers a question of *mechanism*: which propulsion principle does an optimizer select under actuator and stability constraints? Its coefficients are not calibrated, so it is used for direction, not for magnitude;
- *single-leg CFD* of the real BODY2 leg geometry in OpenFOAM with overset grids answers the questions of *magnitude*: how much thrust and power each propulsor produces across forward speed, and where one overtakes the other. It is validated against an experiment and provides every quantitative result of this thesis;
- the *whole-robot* real-time solver Flare Live provides the reference fluke motion. Its fluke force is checked against the CFD and found too low, so its results are used for trends only.

Figure 1.1 shows how the three tools and their results are connected. Where the reduced-order model and the CFD disagree, the CFD is authoritative; the disagreement itself is analysed in Section 9.2.

The scope is limited as follows.

- *Environment*. The robot swims at the surface of still water. Waves, currents and the transition between land and water are excluded. All propulsor CFD is single-phase with a rigid lid; the free surface is not modelled.
- *Motion*. Only steady, periodic, straight-ahead swimming is considered. Turning, starting and feedback control are excluded.
- *Propulsors*. Two propulsors fitted to the same leg are compared: the robot's folding fan paddle (drag-based) and a small swept fluke on a pin joint (lift-based). The fluke pitch is prescribed in the simulations; the passive pin-and-spring realization of the hardware is not simulated.

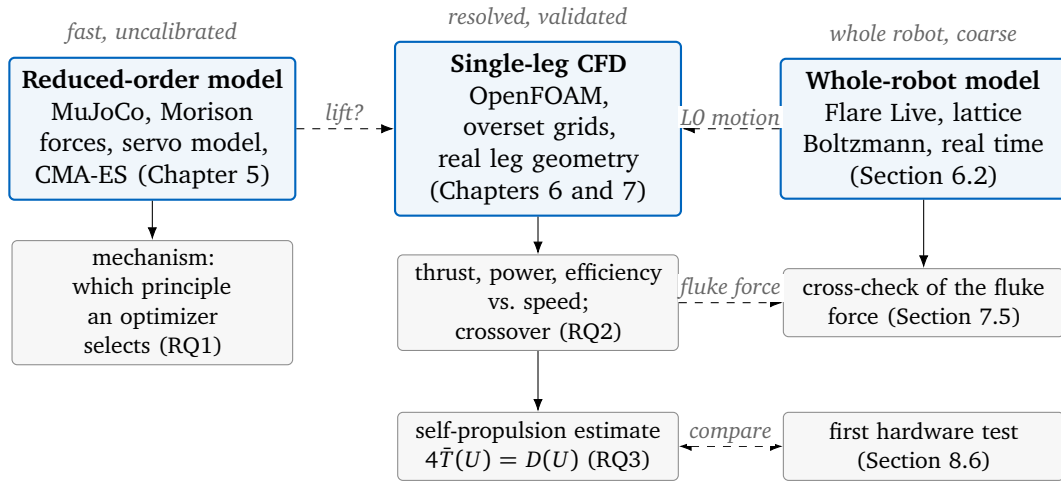


Figure 1.1: Overview of the three simulation tools and their roles. Solid arrows: results; dashed arrows: information passed between tools. The reduced-order model motivates the lift-type propulsor, Flare Live provides the reference fluke motion, and the CFD provides the thrust curves for the self-propulsion estimate and the reference against which the Flare Live fluke force is checked.

- *Evaluation.* Thrust and hydrodynamic input power are computed for the propulsor alone (fan or fluke), without the linkage. Swimming speeds are predicted from a quasi-steady force balance with a range of hull resistances. Hardware experiments are limited to a first timed test from rest.

1.3 Research Questions and Hypotheses

The central question of this thesis is:

Which propulsion principle, drag-based paddling or lift-based flapping, lets the legs of BODY2 swim faster and more efficiently, and how should each propulsor move?

It is split into three research questions, each paired with a hypothesis.

- (RQ1)** *Reduced-order optimization.* When a whole-robot gait is optimized in a reduced-order hydrodynamic model under actuator-speed and attitude limits, which propulsion mechanism does the optimizer select, and does the answer depend on the physics included in the model?

Hypothesis H1. If the model contains a lift term, the optimizer uses lift-based strokes, and at equal speed these need less power than drag-based strokes.

- (RQ2)** *Drag versus lift across speed.* For the real leg geometry, how large are thrust, power and Froude efficiency of an optimized drag-type paddle and of a lift-type fluke between rest and 0.4 m s^{-1} , and what limits each propulsor?

Hypothesis H2. The optimized paddle produces more thrust at low speed, but its thrust falls steeply with speed because the recovery stroke meets an increasing oncoming flow. The fluke produces less thrust at rest, keeps a larger fraction at speed, and becomes both stronger and more efficient above a crossover speed. The fluke performs best when its pitch is adapted to the speed so that the angle of attack stays near the efficient range of flapping foils (about $15 - 25^\circ$).

(RQ3) *Whole robot.* What cruising speed and power follow for the whole robot?

Hypothesis H3. Both optimized propulsors reach similar cruising speeds, set by the hull and linkage resistance, but the fluke needs less power.

RQ1 is answered in Chapter 5. RQ2 and RQ3 are answered in Chapter 8, after the CFD method and its verification in Chapters 6 and 7. Chapter 9 evaluates the hypotheses.

1.4 Contributions

The contributions of this thesis are:

1. *A reduced-order gait-optimization pipeline with an actuator model* (Chapter 5). Morison-type link forces with an optional flat-plate lift term, a DC-motor torque–speed model of the servos and feasibility-first constraints are combined with CMA-ES. In all 101 feasible optima obtained with lift in the model and without a limit on the lift share, lift supplies the forward impulse and drag opposes it. A lift-share limit allows drag-based and lift-based strokes to be compared within the same model.
2. *An optimized drag-type stroke for the BODY2 fan paddle* (Section 8.1). Overset CFD of the real leg with a two-state composite for the folding fan yields a stroke with a trapezoidal velocity law, a 12-mm folded width and early closing that produces 73–83 mN per leg at rest, 3.1–3.5 times the robot’s current gait. The added-mass change that the composite does not resolve is bounded analytically with an added mass identified from the CFD.
3. *A speed-resolved comparison of drag- and lift-type propulsors on the same leg* (Section 8.3). Thrust, power and efficiency are computed from rest to 0.40 m s^{-1} . The drag paddle’s speed limit is traced to its recovery, the thrust and efficiency crossovers are located at about 0.24 and 0.22 m s^{-1} , and a speed-scheduled fluke pitch raises the fluke efficiency at 0.30 m s^{-1} from 0.24 to 0.40 .
4. *Self-propulsion estimates and a fidelity assessment* (Section 8.5 and Chapter 7). Cruising speeds are bounded with a range of hull resistances, the CFD set-up is validated against a flapping-foil experiment and checked for mesh and cycle convergence, and the thrust deficit of the real-time solver Flare Live is quantified on the same fluke motion.

1.5 Thesis Outline

Chapter 2 reviews drag- and lift-based swimming in animals and robots, flapping-foil hydrodynamics and simulation approaches for swimming legged robots. Chapter 3 introduces the quantities and simple models used throughout: quasi-steady paddling, flapping-foil kinematics and the self-propulsion balance. Chapter 4 describes the robot, the two propulsors and their leg motions. Chapter 5 presents the reduced-order model and the gait optimization. Chapter 6 describes the single-leg CFD and the whole-robot Flare Live model, and Chapter 7 verifies and validates the CFD and checks the real-time solver against it. Chapter 8 reports the drag-type stroke design, the lift-type fluke, their comparison, the speed schedule for the fluke pitch and the self-propulsion estimates. Chapter 9 interprets the results and states the limitations, and Chapter 10 concludes and outlines future work. The appendices collect additional data, the coordination results of the reduced-order model (Appendix B), the Flare Live results (Appendix C) and information for reproducing the work.

Chapter 2

Related Work

This chapter reviews what is known about drag- and lift-based swimming in animals and on foils, how swimming legged robots have been designed and simulated, and which gap this thesis addresses.

2.1 Drag- and Lift-Based Swimming in Animals

Fish [Fis96] classified mammalian swimming by the force that produces thrust. Paddlers such as the muskrat, mink or dog move a foot or hand backwards against the water in a power stroke and return it in a recovery stroke. Their propulsive efficiency is at most about 0.33. Swimmers that oscillate a fluke or flipper across the swimming direction generate lift and reach efficiencies above 0.8. Fish argued that the transition from paddling to lift-based swimming in the evolution of aquatic mammals was driven by this efficiency difference and that intermediate forms combine both principles.

The muskrat is the best-documented paddler [Fis84]. Its power stroke lasts 0.18 s and its recovery 0.22 s, a duty factor of 0.45. During recovery it folds the toes together, reducing the frontal area of the foot by 55.5 %, yet the recovery drag still amounts to about one third of the power-stroke thrust. Its maximum mechanical efficiency is 0.33. Dogs use a stereotyped diagonal paddling gait, in which each foot moves on a closed path and the fore- and hindlimbs of opposite sides move together [FDT21]. Ducks paddle with webbed feet whose hydrodynamic performance resembles that of oars rather than delta wings [RG23].

Cetaceans are the reference for lift-based swimming. Bottlenose dolphins swim with a propulsive efficiency of about 0.8 [Fis93]. Odontocetes cruise at Strouhal numbers of 0.26 ± 0.05 [RF04], and flying and swimming animals in general cruise at $0.2 < St < 0.4$, the range of high power efficiency of oscillating foils [TNT03]. Walker and Westneat [WW00] compared rowing and flapping appendages with a blade-element model and concluded that rowing is favourable at low speeds and for manoeuvres and acceleration, whereas flapping is more efficient at higher speeds. For BODY2 this suggests the division of labour tested in this thesis, a paddle for starting and low speed and a fluke for cruising, and makes the speed at which one overtakes the other the quantity to be found.

2.2 Flapping-Foil Hydrodynamics

A foil that heaves and pitches with a phase difference produces thrust through a leading-edge vortex on each half stroke and a reverse von Kármán street in its wake [TTG91]. Wake stability analysis predicts the highest efficiency at $0.25 \lesssim St \lesssim 0.35$ [TTG91]. Anderson et al. [And+98] measured efficiencies up to 0.87 for a NACA 0012 foil at a heave amplitude of 0.75 chord, a maximum angle of attack of about 20° and a pitch lead of 75° . Read et al. [RHT03] extended these measurements and found a broad efficiency plateau of 0.5–0.6 for maximum angles of attack between 15° and 25° and the largest thrust coefficients, around 2.4, at $St \approx 0.6$ and angles of attack of 30° – 35° , where efficiency drops to below 0.5. Hover et al. [HHT04] showed that the time history of the angle of attack matters as much as its maximum: an angle of attack held nearly constant over the stroke extends the efficient range to higher Strouhal numbers and thrust. Schouveiler et al. [SHT05] reported thrust coefficients and efficiencies with error bars over a range of Strouhal numbers and angles of attack; one of their points is used as the validation case in Section 7.1. Floryan et al. [Flo+17] derived scaling laws in which thrust grows with the square of the Strouhal number and efficiency favours large amplitudes at low frequencies. For a self-propelled swimmer, the Strouhal number is not a free parameter but follows from the balance of thrust and drag [Elo12]. Three-dimensional effects reduce thrust at low aspect ratio; cetacean flukes have aspect ratios of about 2–6 [AFM20]. For the fluke of BODY2 these results fix the target: an angle of attack of about 15° – 25° and a Strouhal number of 0.2–0.4, which its pitch must maintain while the swimming speed changes.

2.3 Paddling Mechanics

A paddle produces net thrust only through an asymmetry between power and recovery stroke, either spatial (a smaller area during recovery) or temporal (a slower recovery) [HB24]. Kim and Gharib [KG11] measured the flow around a flat plate performing a paddling rotation and showed that the thrust follows the growth and shedding of the starting vortex rather than the quasi-steady square of the plate velocity. Crustacean-inspired propulsors reach their best performance when the appendage opens quickly at the start of the power stroke and closes early in the recovery [San+26]. A tendon coupling that shortens the transition between the open and folded states of a webbed robotic foot doubled its propulsive efficiency [LGB24]. For beaver-like robots, the hydrodynamics of bendable webbed feet have been modelled with quasi-steady methods and measured in flumes [Che+21; Che+22]. For the fan paddle of BODY2 the levers are therefore the folded area, the timing of the fold and the velocity law of the power stroke; all three are varied in Section 8.1.

2.4 Swimming Legged Robots

Picardi et al. [Pic+23] reviewed underwater legged robots; most walk on the bottom and few swim. Li et al. [LFC+19] analysed the kinematics and hydrodynamics of a robotic dog paddling with rigid limbs. Qu et al. [Qu+25] built an amphibious robotic dog and measured single-leg and whole-robot thrust for paddling gaits with different power-stroke fractions; a shorter power stroke produced more thrust but also larger roll. Wang et al. [Wan+25a] simulated quadrupedal paddling with an immersed-boundary CFD solver. They varied the initial swing angle, the swing amplitude and the power-stroke fraction of rigid, non-folding legs at zero forward speed, and found that the thrust of four legs exceeds four times the single-leg thrust by up to 87%.

Han et al. [Han+25] trained an LSTM model of the leg forces from tank measurements and used it for multi-objective gait optimization. Yao et al. [Yao+23] instead added thrusters to a quadruped and learned a propulsion controller.

Lift-based propulsion has been demonstrated on robots with dedicated flippers. Baines et al. [Bai+22] built a turtle robot with morphing limbs that can paddle and flap with the same limb; the flapping gait produced thrust over most of the stroke and had the lower cost of transport, while paddling gave higher accelerations. The DuckyDog robot uses variable-stiffness legs with passive fins [Wan+25b]. Within the same chair as this thesis, Heber [Heb26] optimized drag-based paddling gaits of an amphibious quadruped by direct collocation over a reduced-order hydrodynamic model calibrated against an SPH simulation. Table 2.1 summarizes these studies. Most measure or simulate thrust at rest, and none compares a drag-based and a lift-based propulsor on the same leg across forward speed, which is what the choice of propulsor for BODY2 requires.

Table 2.1: Swimming legged robots and this thesis. Speeds as reported; [†] derived from the reported course length and time; – not reported.

Study	Propulsor	Method	Thrust vs. forward speed	Swimming speed
Li et al. [LFC+19]	rigid paddling legs (pneumatic soft actuators)	kinematic and drag analysis, prototype	no	–
Qu et al. [Qu+25]	rigid two-segment paddling legs	force measurement at rest, free swimming	no	0.16 m s^{-1} (0.54 BL/s)
Wang et al. [Wan+25a]	rigid paddling legs	immersed-boundary CFD, tethered	no	–
Han et al. [Han+25]	paddling legs with web plates	tank measurements, LSTM model, gait optimization	yes, $0 - 0.3 \text{ m s}^{-1}$	$\approx 0.1 \text{ m s}^{-1\dagger}$
Baines et al. [Bai+22]	morphing limb: paddling and flapping	force measurement at rest, CFD at one speed	no	–
DuckyDog [Wan+25b]	legs with passive fins	free swimming	no	0.30 BL/s
Heber [Heb26]	rigid paddling legs	reduced-order model, direct collocation	no	–
This thesis	folding fan paddle and fluke on the same leg	reduced-order optimization, overset CFD, hardware test	yes, $0 - 0.4 \text{ m s}^{-1}$	CFD $0.26 - 0.30 \text{ m s}^{-1}$ (1.1–1.3 BL/s); hardware 0.14 / 0.26 m s^{-1} mean from rest at 2 / 4 Hz

2.5 Simulation Approaches for Swimming Robots

The simulation methods span several orders of magnitude in cost. Reduced-order models apply quasi-steady drag, added mass and buoyancy to each link [Mor+50; Fos94]. They are fast enough for optimization [Che+21; Heb26; Lee+25], but their coefficients must be identified, and a plain drag model contains no lift. Particle methods such as SPH and grid-based CFD resolve the flow, but one simulated second takes minutes to hours. Overset (Chimera) grids are an established way to simulate large prescribed body motions in OpenFOAM [Wel+98;

Win+20]; the immersed-boundary method is an alternative [Wan+25a]. Lattice-Boltzmann solvers on adaptive block grids, such as the commercial Flare Live [Man26], trade resolution for real-time speed and can simulate a whole robot with its joints and actuators. For this thesis the consequence is the division of tools described in Chapter 1: the reduced-order model for the search and the CFD for the numbers.

2.6 Research Gap

The literature establishes that lift-based flapping is more efficient than drag-based paddling in animals and that paddling can be improved by folding, timing and velocity shaping. Robotic studies of legged swimming have concentrated on drag-based strokes of rigid or folding feet, mostly at zero forward speed [Wan+25a; Qu+25], and on reduced-order models whose lift content is either absent or uncalibrated. To the author’s knowledge, no study has compared an optimized drag-type paddle and a lift-type fluke on the same legged-robot leg across forward speed with a resolved flow simulation, identified what limits each propulsor, and translated the result into swimming speeds of the whole robot. The following combination is therefore missing and is addressed here:

- a reduced-order optimization that includes actuator speed limits and a lift term, to see which principle an optimizer chooses;
- single-leg CFD of the same robot leg with both propulsors, each with an optimized motion, from rest to cruising speed;
- a self-propulsion estimate and a cross-check with a whole-robot real-time solver.

Chapter 3

Fundamentals

This chapter introduces the forces on a moving appendage, two simple models of drag-based and lift-based thrust, the metrics used to compare propulsors and the balance that turns single-leg thrust into a swimming speed. The models are used to interpret the simulations, not to replace them.

3.1 Forces on a Moving Appendage

A rigid appendage that moves through still water with velocity $\mathbf{v}$ relative to the water experiences a pressure and shear force. Its components along and normal to $\mathbf{v}$ are the drag and the lift. For a thin plate the drag depends strongly on the orientation: moved broadside, a plate has a drag coefficient of about 1.2–2 [New77]; moved edgewise, it has only skin friction. Accelerating the plate also accelerates the surrounding water, which appears as an added-mass force proportional to the acceleration. Reduced-order models sum these contributions per body [Mor+50; Fos94]:

$$\mathbf{F} = -\frac{1}{2}\rho C_D A |\mathbf{v}| \mathbf{v} + \mathbf{L} - C_a \rho V \dot{\mathbf{v}} + \mathbf{F}_b, \quad (3.1)$$

with density ρ , reference area A , volume V , added-mass coefficient C_a and buoyancy $\mathbf{F}_b$. A pure drag model ($\mathbf{L} = \mathbf{0}$) can only push water backwards; it is the model of a *paddle*. A lift term normal to $\mathbf{v}$ makes it possible to produce thrust with a foil that moves mostly across the swimming direction; it is the model of a *fin* or *fluke*.

The flows of this thesis are unsteady and separated. With a paddle-tip speed of about 0.35 m s^{-1} and a fan radius of 42 mm, the paddle works at a Reynolds number $Re = vc/\nu \approx 1.5 \times 10^4$. The fluke, with a chord of 34 mm at 0.22 m s^{-1} , works at $Re \approx 7.6 \times 10^3$. At these Reynolds numbers a starting paddle forms a vortex whose growth sets the force [KG11], and a flapping foil carries a leading-edge vortex [TTG91; And+98]. Equation (3.1) with constant coefficients misses both effects. It is used here to derive trends, while magnitudes come from CFD.

3.2 Quasi-Steady Paddling

Consider a paddle that sweeps a path of length L backwards (power stroke) with speed v_p and area A , and returns (recovery stroke) with speed $v_r = x v_p$ and a reduced area aA , $0 < a \leq 1$. At zero forward speed the quasi-steady impulses of the two strokes are $\frac{1}{2}\rho C_D A v_p L$ and $-\frac{1}{2}\rho C_D a A x v_p L$, and the period is $L(1+x)/(x v_p)$. The cycle-mean thrust is

$$\bar{T} = \frac{1}{2}\rho C_D A v_p^2 \frac{x(1-ax)}{1+x}. \quad (3.2)$$

Both asymmetries identified by Herrera-Amaya and Byron [HB24] appear: a spatial one ($a < 1$) and a temporal one ($x \neq 1$). For a fixed power-stroke speed, which is the natural constraint of a speed-limited servo, (3.2) is maximal at

$$x^* = \sqrt{1+1/a} - 1, \quad \text{DUTY}^* = \frac{x^*}{1+x^*}, \quad (3.3)$$

where DUTY is the fraction of the cycle spent in the power stroke. A well-folding paddle (small a) should return quickly, a poorly folding one slowly. For the muskrat, whose foot area is reduced by 55.5 % during recovery ($a = 0.445$), (3.3) gives $\text{DUTY}^* = 0.445$, close to the measured 0.45 [Fis84]. For BODY2's fan paddle the effective ratio a follows from the CFD (Section 8.1); when $x^* > 1$, the recovery would have to be faster than the power stroke, and the servo speed limit then sets DUTY instead of (3.3).

Three further conclusions follow from the same model.

- *Velocity law.* For a given peak speed and stroke time, a trapezoidal velocity profile (short ramps, constant speed in between) produces a larger impulse $\int v^2 dt$ than a sinusoidal one. For a given impulse, Hölder's inequality shows that the energy $\int v^3 dt$ is minimal when v is constant.
- *Forward speed.* When the robot moves with speed U , the power stroke acts on the relative velocity $v_p - U$ and the recovery on $v_r + U$. The thrust of the power stroke falls and the drag of the recovery rises with U . The paddle should be open only while $v > U$.
- *Efficiency.* The useful power is $\bar{T}U$, while the paddle delivers $\bar{T}v_p$ to the water during the power stroke. The Froude efficiency of a drag-based stroke is therefore bounded by $U/v_p < 1$ and is highest for a large paddle moved slowly [Fis96; WW00].

3.3 Flapping-Foil Kinematics

A foil that is carried with forward speed U and heaves with vertical position $h(t)$ meets the water at the inflow angle

$$\gamma(t) = \arctan \frac{\dot{h}(t)}{U}. \quad (3.4)$$

If the chord is pitched by $\theta(t)$ from the swimming direction, the effective angle of attack is

$$\alpha(t) = \gamma(t) - \theta(t), \quad (3.5)$$

with signs chosen so that θ and γ rotate the same way (Figure 3.1). The lift L acts normal to the relative inflow and the drag D along it; the thrust is

$$F_x = L \sin \gamma - D \cos \gamma. \quad (3.6)$$

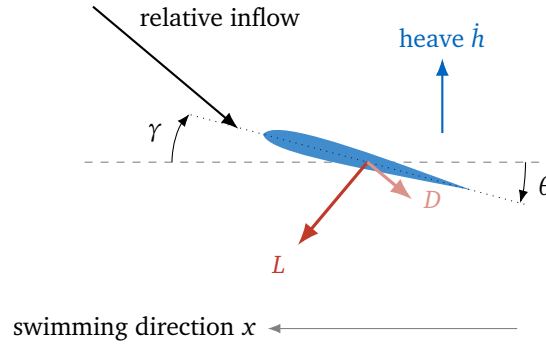


Figure 3.1: Angles of a heaving and pitching foil. The relative inflow is tilted by the inflow angle $\gamma = \arctan(\dot{h}/U)$ below the swimming direction; the chord is pitched by θ in the same sense; the angle of attack between chord and inflow is $\alpha = \gamma - \theta$. Lift L acts normal and drag D parallel to the relative inflow; the thrust is their net component along x . Shown for the upstroke.

Thrust is produced as long as the lift is tilted forwards enough to overcome the drag. Because $\sin \gamma$ multiplies the lift, the same foil at the same α produces a forward force whenever it heaves; unlike the paddle, it does not need a recovery stroke against the flow.

The Strouhal number $St = fA/U$, with the peak-to-peak excursion A of the trailing edge, is the ratio of heave speed to forward speed. In this thesis A is taken as the excursion of the pin, which is known from the prescribed motion; the trailing-edge value is somewhat larger. Efficient cruising lies at $0.2 < St < 0.4$ [TNT03; TTG91]. A leg driven by a speed-limited servo has an approximately fixed heave speed $\dot{h}_{\max}$. The Strouhal number, and with it the inflow angle, then decreases with swimming speed:

$$\gamma_{\max}(U) = \arctan \frac{\dot{h}_{\max}}{U}. \quad (3.7)$$

If the pitch schedule $\theta(t)$ is kept fixed while U changes, the angle of attack changes as well. At rest ($U \rightarrow 0$) the inflow is vertical, $\gamma = 90^\circ$, and the fluke is stalled; it works as a small paddle. At high speed $\gamma_{\max}$ becomes small and a fixed pitch amplitude can make α too small or even negative.

A simple way to keep α in the efficient range of $15-25^\circ$ [RHT03; HHT04] is to scale the pitch with the inflow angle,

$$\theta(t) = \theta_0 \frac{\dot{h}(t)}{\dot{h}_{\max}}, \quad \theta_0 = \frac{1}{2} \gamma_{\max}(U), \quad (3.8)$$

so that at mid-stroke, where $\dot{h} = \dot{h}_{\max}$, the foil takes half of the inflow angle as pitch and the other half as angle of attack. This *speed-scheduled pitch* is evaluated in Section 8.4. It requires one parameter per speed and no knowledge of the flow. For BODY2's leg, $\dot{h}_{\max} \approx 0.28 \text{ m s}^{-1}$, so $\gamma_{\max} = 52^\circ, 43^\circ$ and 35° at 0.223 m s^{-1} , 0.30 m s^{-1} and 0.40 m s^{-1} .

3.4 Performance Metrics

All single-leg quantities are cycle means over complete stroke periods $T = 1/f$. For a propulsor with position $\mathbf{x}_H(t)$ of its reference point, angular velocity $\boldsymbol{\omega}(t)$ and fluid force and moment $\mathbf{F}$,

$\mathbf{M}_H$ about that point:

$$\bar{T} = \langle F_x \rangle, \quad (3.9)$$

$$\bar{P} = -\langle \mathbf{F} \cdot \dot{\mathbf{x}}_H + \mathbf{M}_H \cdot \boldsymbol{\omega} \rangle, \quad (3.10)$$

$$\eta = \frac{\bar{T} U}{\bar{P}}, \quad C_T = \frac{\bar{T}}{\frac{1}{2} \rho U^2 S}. \quad (3.11)$$

The x axis points in the swimming direction, so that $\bar{T} > 0$ is thrust and $\bar{P} > 0$ is power delivered to the fluid. $\bar{P}$ is the hydrodynamic input power: it does not include the inertia of the parts, friction or motor losses. η is the Froude (propulsive) efficiency; it vanishes at $U = 0$, where the thrust per power $\bar{T}/\bar{P}$ is reported instead. S is the planform area of the propulsor. For the whole robot the cost of transport is reported either as energy per distance $\bar{P}/U$ (J/m) or dimensionless as $\bar{P}/(mgU)$; the form used is stated with each value.

For the drag-type paddle, the thrust is also split into the contributions of the power stroke and the recovery,

$$\bar{T} = \bar{T}_{\text{pow}} + \bar{T}_{\text{rec}}, \quad \bar{T}_{\text{pow}} = \frac{1}{T} \int_{\text{power}} F_x dt, \quad \bar{T}_{\text{rec}} = \frac{1}{T} \int_{\text{recovery}} F_x dt, \quad (3.12)$$

which shows directly which half of the cycle limits the thrust at speed.

3.5 Self-Propulsion Balance

A robot swims steadily at the speed where the propulsive force of its legs equals the resistance of the hull and the parts that do not propel. If each of the four legs produces the cycle-mean thrust $\bar{T}(U)$ measured in isolation and the legs do not interact, the cruising speed U_c solves

$$4 \bar{T}(U_c) = D(U_c), \quad (3.13)$$

and the time history from rest follows from the quasi-steady surge equation

$$(m + m_a) \dot{U} = 4 \bar{T}(U) - D(U). \quad (3.14)$$

Equation (3.13) neglects the interaction between the legs, which can raise the thrust of four paddling legs above four times the single-leg value [Wan+25a], and the effect of the leg motion on the hull resistance. It also assumes that the legs keep their prescribed motion at every speed. Because the resistance of BODY2 is known only within bounds, (3.13) is solved for three resistance curves in Section 8.5, and the result is reported as a range.

Chapter 4

Robot Platform and Propulsors

This chapter describes the quadruped BODY2, the two propulsors that are compared and the leg motions that drive them. The motions defined here are the inputs of the CFD in Chapters 6 and 8.

4.1 The Quadruped BODY2

BODY2 is a small amphibious quadruped built at the chair. Its sealed, flat-bottomed hull carries the electronics and the eight servos and gives the robot its buoyancy (Figure 4.1). At the design waterline the hull is 232 mm long and 99 mm wide, with a draft of 25 mm and a displacement of about 385 g. The complete robot weighs 418.8 g and therefore floats 1–2 mm deeper, consistent with the waterline observed on the hardware. The robot swims at the surface: the hull floats and the legs reach down into the water.

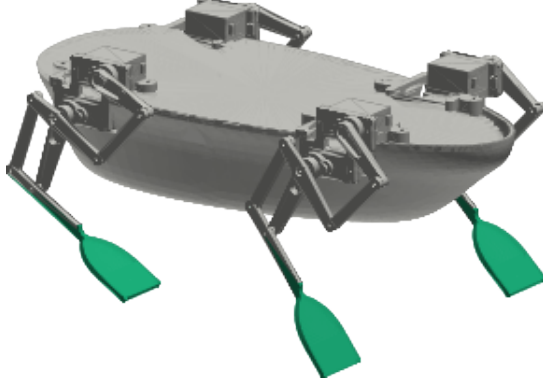
Each leg is a planar linkage with two degrees of freedom that moves in a vertical plane parallel to the hull's symmetry plane. Two servos (PTK 7465W; stall torque 0.44–0.57 N m and no-load speed $650–860^\circ\text{s}^{-1}$ at 6.0–8.4 V) drive it. The first rotates a crank about the hip axis O_2 (angle ψ); the second sets the configuration of a parallelogram that carries the foot (angle ϕ). The end point E of the parallelogram carries the propulsor. For walking, the foot is the fan paddle described below. For swimming, the leg can be fitted either with this paddle or with a fluke on the same mounting points, so that the two propulsion principles can be compared on one leg.

4.2 Propulsors

Figure 4.2 and Table 4.1 summarize the two propulsors.

Drag-type: folding fan paddle. The foot is a 120° fan of radius 42 mm at the end of a 50 mm rod; *fan* denotes the folding blade alone and *paddle* the blade on its rod. Open, the fan has an area of 18.5 cm^2 (20.5 cm^2 including the rod). A cable folds it for the recovery stroke. In the current hardware the folded fan is still 30 mm wide (12.5 cm^2 , 68 % of the open area), which

(a) Drag-type configuration (fan paddles)



(b) Lift-type configuration (flukes)

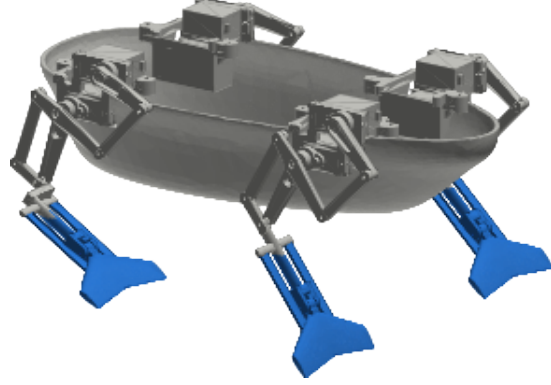
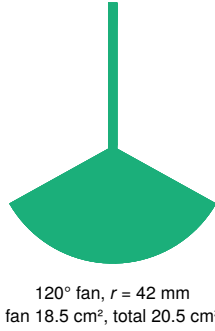
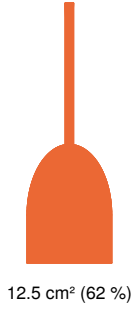


Figure 4.1: BODY2 in its two swimming configurations, rendered from the CAD models. (a) Drag-type configuration with the folding fan paddles (green) used for walking and paddling. (b) Lift-type configuration with a fluke (blue) on a straight foot rod at each leg.

(a) Fan paddle, open



(b) Closed, 30 mm



(c) Closed, 12 mm



(d) A2 fluke (top view)

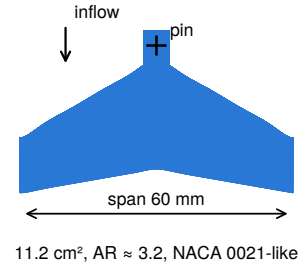


Figure 4.2: Propulsor geometries used in the CFD, drawn to scale. (a) Open fan paddle. (b) Folded fan of the current hardware, 30 mm wide. (c) Folded fan of 12 mm width assumed for the optimized stroke. (d) Fluke A2 in top view, with its pin axis.

makes the spatial asymmetry weak. A folded width of 12 mm (5.9 cm²) is assumed for the optimized stroke in Section 8.1. The fan is always oriented normal to the stroke; its opening and closing are treated as instantaneous switches between two rigid shapes (Section 6.1.4).

Lift-type: fluke. The fluke (design “A2”) is a small swept foil with a span of 60 mm, a root chord of 34 mm, a planform area of 11.2 cm² and an aspect ratio of about 3.2. Its section is a thick symmetric profile similar to NACA 0021. It is mounted on a pin near its leading edge at the lower end of a straight foot rod, so that it can pitch about a spanwise axis. On the hardware the pitch is passive, set by the hydrodynamic moment against a leaf spring and two end stops. In the simulations of this thesis the pitch angle $\theta(t)$ is prescribed; how closely a passive fluke follows a prescribed schedule is not studied here.

4.3 Leg Motions

Figure 4.3 shows the motions studied.

Table 4.1: Propulsor data. Areas are planform areas of the propulsor without the linkage.

	Fan paddle (drag type)	Fluke A2 (lift type)
Shape	120° fan, $r = 42$ mm, on 50 mm rod	swept foil, NACA 0021-like
Area	18.5 cm ² open; 12.5 or 5.9 cm ² folded (30 or 12 mm wide)	11.2 cm ²
Size	fan radius 42 mm	chord 34 mm, span 60 mm
Principle	drag, folding recovery	lift, pitching about pin
Degrees of freedom	ψ, ϕ + open/closed	ψ, ϕ + pitch θ
Frequency	0.8 Hz (robot gait), 1.39 Hz (V3trap)	2 Hz

Robot gait. The robot currently swims with a gait taken from video of the robot: period 1.25 s, DUTY = 0.5, a sinusoidal hip sweep of 60° and the fan folded to 30 mm during the recovery. It was designed for walking and serves as the baseline.

Optimized drag-type stroke (V3trap). The stroke derived in Section 8.1 has a period of 0.72 s and DUTY = 0.382. The hip sweeps 50° centred on the vertical (120–70°), so that the paddle moves nearly straight backwards and stays normal to its path. The hip velocity follows a trapezoidal law with ramps of about 45 ms. During the recovery the parallelogram curls the paddle upwards and the fan is folded to 12 mm. The fan is closed at $\tau_c = 0.33$, the start of the power-stroke deceleration, and opened at $\tau_o = 0$, the reversal point at the end of the recovery (Figure 4.3a).

Fluke motion. The fluke leg keeps the hip fixed and swings the foot rod about the knee, so that the fluke's pin moves up and down on a flat arc: $\pm 24.9^\circ$ at 2 Hz, a vertical stroke of 39.2 mm and a peak heave speed of about 0.28 m s^{-1} (Figure 4.3b). The reference motion, called L0, is taken from the whole-robot Flare Live model (Section 6.2): the joint angles and the fluke pitch of one leg were recorded while the fluke pitch was controlled to hold a target angle of attack at 0.223 m s^{-1} , and phase-averaged over the recorded cycles. The resulting 75th percentile of $|\alpha|$ at this speed is 19.9° . Three families are derived from it:

- *L2, fixed pitch schedule:* the L0 motion unchanged at 0 m s^{-1} , 0.08 m s^{-1} , 0.15 m s^{-1} and 0.30 m s^{-1} , as a propeller is tested at fixed rotational speed and varying advance speed. Since γ changes with U (Equation (3.7)), the angle of attack is large at low speed and small at high speed.
- *L1, angle-of-attack law:* a pitch schedule computed from the heave speed to hold a target α of 10° or 28° at 0.223 m s^{-1} , and 20° at 0.40 m s^{-1} . These cases use slightly different hinge paths from L0 and are therefore not a clean angle-of-attack sweep.
- *L3, speed-scheduled pitch:* the L0 hinge path with the pitch schedule (3.8), $\theta_0 = \gamma_{\max}/2$, at 0.30 m s^{-1} and 0.40 m s^{-1} ($\theta_0 = 21.6^\circ$ and 17.6°).

Two differences between the propulsors are inherent in this set-up and are kept in mind throughout. First, the propulsors differ in area (18.5 versus 11.2 cm^2) and in frequency, so thrust per leg, thrust per area and thrust per power are all reported. Second, the drag-type stroke has been optimized in several rounds, whereas L0 is a motion taken from another model and was not optimized for thrust; L3 is the only systematic improvement of the fluke motion. At rest both propulsors draw a similar hydrodynamic input power of 18–22 mW per leg, so that the comparison is made at a similar power budget.

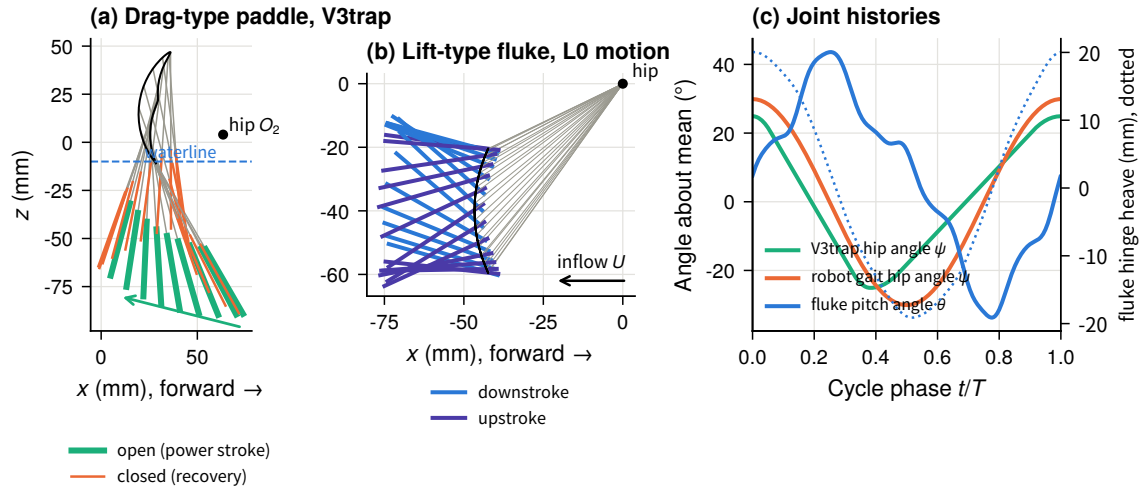


Figure 4.3: Leg motions in the hull frame. (a) Optimized drag-type stroke V3trap: paddle positions at 25 instants per cycle, open (thick) and closed (thin), and the path of the linkage end point E . (b) Fluke motion L0: chord positions over one cycle, downstroke and upstroke, with the hinge path. (c) Joint histories: hip angle of V3trap and of the robot gait (about their means), fluke pitch angle and hinge heave of L0 (dotted, right axis).

4.4 Hardware Prototype

The lift-type configuration exists as hardware (Figure 4.4). All four legs carry printed flukes on a passive pin with a leaf spring. The firmware drives each leg through seven control points per cycle: the thigh is held fixed, the foot bar swings by $\pm 25^\circ$ about the horizontal, and the legs are coordinated in a trot. At 2 Hz this is close to the L0 motion: the pin travels 38.0 mm vertically (L0: 39.2 mm) at a mean heave speed of about 0.15 m s^{-1} (L0: 0.157 m s^{-1}), but its peak heave speed is about 15 % lower, because the control points are joined by straight segments. Unlike in the simulations, the hull is mounted reversed, so the robot swims with its blunt end first. In the water, BODY2 with flukes swam visibly faster than with its fan paddles and its current gait. A first timed test is compared with the self-propulsion estimate in Section 8.6. The optimized paddle was not tested, because it requires a controllable fold and a trapezoidal velocity law that the hardware does not have.

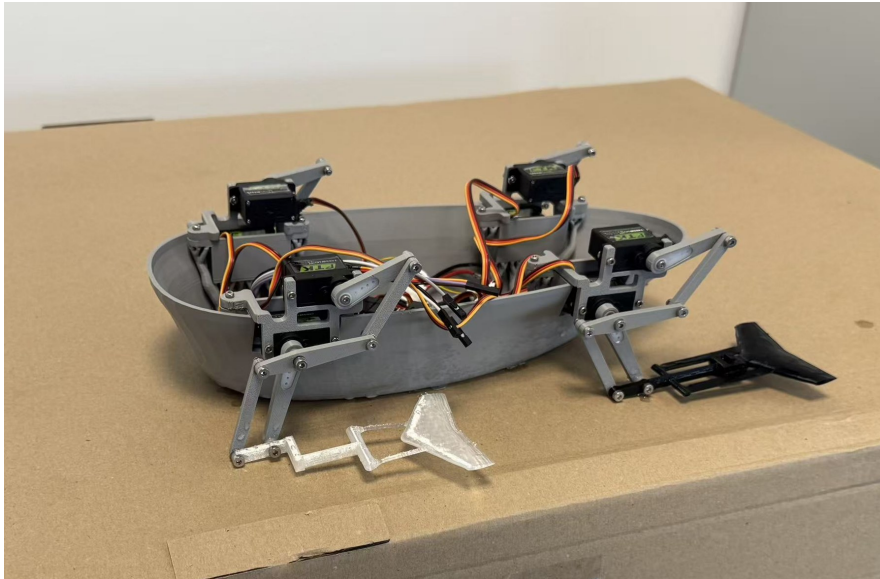


Figure 4.4: Hardware prototype of BODY2 with printed fluke feet mounted on the leg linkages; two of the fluke feet are shown resting on the table. Each leg is driven by two PTK 7465W servos.

Chapter 5

Reduced-Order Gait Optimization

This chapter addresses RQ1. A whole-robot model with quasi-steady link forces is built in MuJoCo [TET12], and its gait is optimized with CMA-ES [HO01; Han16] under actuator and attitude constraints. The model is cheap enough for thousands of evaluations but its coefficients are not calibrated. Its purpose is to show which propulsion principle an optimizer selects, and how the answer depends on the physics included, not to predict speeds of BODY2.

5.1 Model

5.1.1 Robot

BODY2 could not be used directly: its CAD-derived model is an open kinematic tree in which the parallelogram linkages are not closed, and its buoyancy distribution had not been measured. The optimization therefore uses a simple quadruped model, `toy_quad`, of similar size (Figure 5.1, Table 5.1). Each leg has three hinge joints about the transverse axis: hip, knee and flipper wrist, twelve actuators in total. The flipper, a thin plate of 52 mm $\times$ 36 mm whose area of 18.7 cm² is close to that of the open fan paddle of BODY2, is called *flipper* throughout to distinguish it from the fan paddle and the fluke of the real robot: it can paddle or flap depending on the physics included, whereas the two propulsors of BODY2 are each built for one principle. The dry mass is 0.78 kg; fully submerged, the robot would displace 1126 cm³, so it floats at the surface like BODY2. The body length of 0.239 m is used to express speeds in body lengths per second.

5.1.2 Hydrodynamic Forces

Each geometric body k receives a quasi-steady force, scaled by its immersed fraction f_k and applied at its centre:

$$\mathbf{F}_k = \rho g V_k f_k \mathbf{e}_z - \frac{1}{2} \rho f_k \sum_{i=1}^3 C_{D,i} A_i v_i |v_i| \mathbf{e}_i - c_v f_k \mathbf{v} + \mathbf{L}_k. \quad (5.1)$$

Here $\mathbf{v}$ is the velocity of the body centre relative to still water, v_i its components along the body axes $\mathbf{e}_i$, and A_i the projected areas; c_v is a small linear damping coefficient (0.05, 0.02



Figure 5.1: The MuJoCo model `toy_quad` floating at the surface (rendering from the simulation). Each leg has a hip, a knee and a flipper joint.

Table 5.1: The reduced-order robot model `toy_quad`.

Part	Geometry	Size	Mass
Trunk	box	$0.20 \text{ m} \times 0.10 \text{ m} \times 0.05 \text{ m}$	0.600 kg
Upper leg ($\times 4$)	capsule	$r = 7 \text{ mm}, l = 55 \text{ mm}$	0.020 kg
Lower leg ($\times 4$)	capsule	$r = 6 \text{ mm}, l = 50 \text{ mm}$	0.015 kg
Flipper ($\times 4$)	plate	$8 \text{ mm} \times 52 \text{ mm} \times 36 \text{ mm}$	0.010 kg
Joint ranges: hip and knee $\pm 1.3 \text{ rad}$, flipper $\pm 1.6 \text{ rad}$; legs at $\pm 0.085 \text{ m}$ fore-aft, $\pm 0.045 \text{ m}$ lateral Dry mass 0.78 kg; added mass 0.28 kg; body length 0.239 m; time step 2 ms, implicit integrator			

and 0.10 N s m^{-1} for flipper, links and trunk) that matters only at very low speed. The drag is anisotropic: the flipper has $C_D = (2.2, 0.10, 0.10)$ normal to and along its faces, the links $(0.5, 0.5, 0.25)$ and the trunk $(0.9, 1.6, 1.8)$. This anisotropy is the only source of thrust in a drag-only model. The immersed fraction is computed from the actual vertical extent of each body, so a flipper that breaks the surface loses its force. Added mass $m_a = C_a \rho V$ ($C_a = 1.0$ for the flipper, 0.3 for the links, 0.2 for the trunk) is added to the body inertia together with a constant upward force $m_a g$ that cancels its weight. This isotropic approximation is unconditionally stable, unlike an explicit $\dot{v}$ term. The coefficients are literature values of the same level as in beaver-robot models [Che+21; Che+22]; they are not calibrated.

Lift. All terms of (5.1) except L_k oppose some component of the relative flow. An optional post-stall flat-plate lift acts on the flippers,

$$C_L(\alpha) = c_l \sin 2\alpha, \quad |L| = \frac{1}{2} \rho C_L A_{\text{face}} |\mathbf{v}|^2 f, \quad (5.2)$$

normal to $\mathbf{v}$ in the plane spanned by $\mathbf{v}$ and the plate normal, where α is the angle between $\mathbf{v}$ and the plate. The value $c_l = 1.1$ is a textbook flat-plate value without an aspect-ratio correction. At small α the lift grows like α and the drag like α^2 , so a fast, shallow, feathered stroke produces almost no thrust in the drag-only model but a large thrust with lift. With the lift switched off, a lift-based gait cannot win, because the mechanism that would make it win is absent.

Thrust decomposition. To identify the mechanism of a gait, the signed impulses of the lift terms and of the drag terms (including the linear term) along the swimming direction are

accumulated over the scoring window. A gait with positive lift impulse and negative drag impulse is *lift-propelled*: lift supplies the forward impulse, and drag only opposes it. The decomposition was checked with a coasting test, in which the initial momentum equals the sum of the impulses. To quantify how much of the propulsion lift provides, the *lift thrust share*

$$s_L = \frac{I_L}{|I_{\text{trunk}}|} \quad (5.3)$$

relates the forward impulse I_L of the lift terms to the drag impulse of the trunk; the drag on the limbs is accumulated separately. A paddling stroke has $s_L \approx 0$; a lift-propelled stroke has $s_L > 1$, because lift must also overcome the drag of the limbs themselves. The share can be imposed as a further limit, $s_L \leq s_{\text{max}}$, which forces drag-based strokes in a model that contains lift (Section 5.4.3).

5.1.3 Servo Model

The actuators are position servos with a stiffness of 60 N m rad^{-1} and critical damping. Real hobby servos cannot move arbitrarily fast. Each actuator is therefore given the torque–speed characteristic of a DC motor,

$$\tau = \tau_s u - \frac{\tau_s}{\omega_{\text{max}}} \omega, \quad |u| \leq 1, \quad (5.4)$$

with stall torque $\tau_s = 3 \text{ N m}$ and no-load speed $\omega_{\text{max}} = 400^\circ \text{ s}^{-1}$. These values were chosen before the servos of BODY2 were identified; the PTK 7465W has a lower stall torque and a higher no-load speed (Section 4.1), and the consequences are discussed in Section 9.4. The back-EMF term is a viscous damping $b = \tau_s / \omega_{\text{max}} \approx 0.43 \text{ N m s rad}^{-1}$, which is added to the joint damping and integrated implicitly by MuJoCo; the result is independent of the time step. Two simpler variants failed: limiting only the rate of the command let the legs be dragged back and then snap forward at $1500^\circ \text{ s}^{-1}$, and rewriting the torque limit at every step acted as an explicit damping that oscillated at up to $3000^\circ \text{ s}^{-1}$ and changed the result six-fold with the time step.

Two power measures are reported. The optimization uses the mean mechanical power at the motor shafts, $\sum_j \langle |\tau_{m,j} \omega_j| \rangle$ with the motor torque $\tau_m = \tau - b \omega$ and no recovery of braking energy. The electrical power adds the copper loss of the same motor model, $\sum_j \langle \max(\tau_{m,j} \omega_j, 0) + \tau_{m,j}^2 \omega_{\text{max}} / \tau_s \rangle$, which follows from (5.4) without new parameters; it is computed after the optimization and used as a check. The cost of transport in this chapter is power over speed in J m^{-1} .

5.2 Gait Parametrization

Every actuator i follows the target angle

$$q_i(t) = q_{0,i} + r(t) \sum_{h=1}^H A_{i,h} \sin(h \vartheta(t) + \varphi_{i,h}), \quad H = 2, \quad (5.5)$$

with offset $q_{0,i}$, amplitudes $A_{i,h}$, phases $\varphi_{i,h}$ and a ramp $r(t)$ from 0 to 1 over the first 1.5 s. The duty warp $\vartheta(t)$ lets the power stroke be faster or slower than the recovery while keeping the period: with the cycle progress $c = (f t) \bmod 1$,

$$\vartheta = \begin{cases} \pi c / \text{DUTY}, & c < \text{DUTY}, \\ \pi + \pi(c - \text{DUTY}) / (1 - \text{DUTY}), & c \geq \text{DUTY}. \end{cases} \quad (5.6)$$

The second harmonic, which is invariant under a phase shift of π , can express the feathering of the flipper. It is not necessary for net thrust: a single harmonic already propels the model. Joints of the same type share amplitude and offset. The search ranges are $f \in [0.2, 2.5]$ Hz, $DUTY \in [0.25, 0.75]$, $A \in [0, 0.9]$ rad and $q_0 \in [-0.5, 0.5]$ rad, normalized to $[0, 1]$.

The coordination of the legs is either free, with an independent phase for every actuator (35 parameters), or fixed to a *gait family* with one phase per joint type (17 parameters). The families are defined by the delay of each leg relative to the left fore leg FL, applied in time before the duty warp:

Family	Common name	Delays of FR, BL, BR (fraction of period)
diag	trot	0.5, 0.5, 0
lr	pace	0.5, 0, 0.5
fb	bound	0, 0.5, 0.5
wave	walk	0.5, 0.75, 0.25
inphase	pronk	0, 0, 0

To classify a free gait, the delays of the joint type with the largest first-harmonic amplitude are compared with each family, and the family with the smallest mean circular distance is reported together with that distance (0 = identical, 0.5 = opposite).

5.3 Objective, Constraints and Optimizer

Each evaluation settles the robot for 1.0 s, ramps the gait in over 1.5 s and scores the following 8 s, rounded up to whole stroke periods. The speed is the displacement along the initial heading divided by the window length, so turning is penalized automatically. Two objectives are used: the highest speed, or the lowest mean mechanical power subject to a minimum speed $v_{\min}$. All settings are listed in Table D.1.

Stability is imposed by limits, not by weights. With RMS values of heading error, roll and pitch and their limits ℓ_k , the penalty is

$$p = \sum_k \frac{\max(0, \text{RMS}_k - \ell_k)}{\ell_k} \left[+ \text{surfacing and } v_{\min} \text{ terms} \right], \quad \text{fitness} = \begin{cases} \text{objective}, & p = 0, \\ -1000 - p, & p > 0. \end{cases} \quad (5.7)$$

Any infeasible gait thus scores below every feasible one; speed cannot buy back a violation. An earlier additive weighting of speed and attitude failed once the legs could move freely: speed dominated, and the best gait rolled by 42° and yawed by 44° . Two limit sets are used: a moderate one, heading/roll/pitch $10/10/15^\circ$, used in this chapter, and a strict one, $10/5/5^\circ$, used for the coordination study in Appendix B. Where stated, the fraction of time in which any limb is less than half immersed (surfacing) is limited to 5 %, because the model has no free-surface physics.

The optimizer is CMA-ES with population 16 and initial step size 0.25 in the normalized space, run for 2500 evaluations unless stated otherwise, with several seeds per condition. Each result is a bound of what the model allows: a lower bound for the highest speed and an upper bound for the least power. Values are given as mean $\pm$ standard deviation over seeds.

Table 5.2: Least mean mechanical power to reach a required speed, moderate limits, best over five families and two seeds (family of the best solution and number of feasible searches out of ten in brackets). Left: drag-based strokes in a model without lift; middle: drag-based strokes forced by $s_L \leq 0.1$ in the model with lift (Section 5.4.3); right: unconstrained strokes in the model with lift.

Required speed	Drag-only model	Lift model, $s_L \leq 0.1$	Lift model, free
0.10 m s^{-1}	0.23 W (walk)	0.42 W (bound; 2/10)	0.10 W (bound; 10/10)
0.20 m s^{-1}	1.04 W (bound)	5.50 W (pronk; 1/10)	0.55 W (pronk; 8/10)
0.30 m s^{-1}	3.10 W (trot)	none (0/10)	0.97 W (bound; 4/10)

5.4 Results

5.4.1 Baseline and Servo Limit

A hand-designed trot at 1.2 Hz swims at 0.070 m s^{-1} (0.29 body lengths per second) with 26.7° RMS pitch and servos saturated 85 % of the time. Without the servo model, the optimizer found gaits with joint speeds of $2700\text{--}4200^\circ \text{ s}^{-1}$, about seven to ten times the servo limit. Replayed with the servo model, the best of them did not move forward (-0.001 m s^{-1}). Re-optimized with the servo model, three free-phase searches reached $0.11\text{--}0.25 \text{ m s}^{-1}$, all close to a bound. The servo limit therefore has to be part of the model; all following results include it.

5.4.2 Drag versus Lift at Equal Speed

Comparing the fastest gaits of the two physics would be unfair, because power rises steeply with speed and the fastest gaits differ in speed. The comparison is therefore made at equal speed: the least mechanical power needed to reach $v_{\min} \in \{0.10, 0.20, 0.30\} \text{ m/s}$ under moderate limits and at most 5 % surfacing, over five families and two seeds (Table 5.2, Figure 5.2).

Three results are robust. First, whenever lift is in the model and not suppressed, every feasible optimum is lift-propelled: in all 22 feasible lift optima of this experiment the lift impulse is positive (0.46–2.2 N s on the front) and the drag impulse negative. The optimizer never paddles when it can flap. Second, at equal speed the lift-propelled gaits need about one half to one third of the power of the best drag-only gaits, at all three speeds. Third, each optimum depends on its physics: with lift removed, the lift optima keep only 35–58 % of their speed, and 26 of the 27 drag optima no longer satisfy the attitude limits when lift is added, because lift also creates moments on the trunk (Section 5.4.3). The size of the ratio is not reliable: the searches have not converged (a second seed lowered the drag-only power at 0.20 m s^{-1} from 2.99 to 1.04 W), so both fronts are upper bounds. The ranking is the same with the electrical power, which includes the copper losses of the servos: the drag-only front needs 0.22, 1.14 and 2.80 W, the lift front 0.10, 0.47 and 0.86 W.

5.4.3 Drag-Based Strokes within the Lift Model

The comparison of Table 5.2 between the drag-only model and the lift model is a comparison of two models as much as of two propulsion principles. A direct test of RQ1 compares both principles within one model. For this, the least-power searches of the lift model were repeated with the additional limit $s_L \leq 0.1$ from (5.3), which admits only strokes that obtain their thrust from drag, with the same speeds, limits, families and seeds as the unconstrained searches (30 searches).

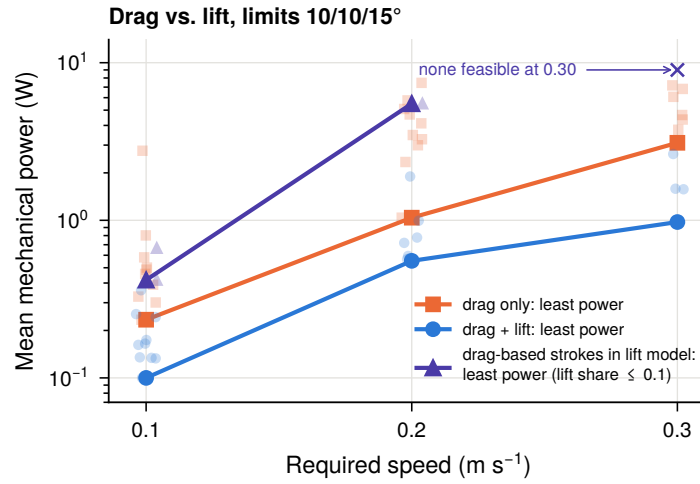


Figure 5.2: Least mechanical power to reach a required speed, moderate limits: drag-only model, model with lift, and drag-based strokes forced by $s_L \leq 0.1$ in the model with lift. Pale markers are the individual optima of each family and seed.

The results are shown in the middle column of Table 5.2 and in Figure 5.2. Only 3 of the 30 searches found a feasible drag-based stroke; most of the others satisfy the share limit but not the required speed. On this small basis, the drag-based strokes need 4.2 times the power of the lift-propelled ones at 0.10 m s^{-1} (0.42 against 0.10 W) and 10 times at 0.20 m s^{-1} (5.50 against 0.55 W), and none reaches 0.30 m s^{-1} , the closest reaching 0.26 m s^{-1} . Because the feasible region is narrow, part of the factor may be due to search difficulty rather than physics; with two seeds and 2500 evaluations, the direction is reliable but the value is not.

The two comparisons bound the answer. In the drag-only model a paddling stroke does not have to suppress the lift that its flipper would produce, which favours it; in the lift model it has to, and the constrained search is difficult, which penalizes it. Together they indicate that in this model a drag-based stroke needs between about two and ten times the power of a lift-based stroke at the same speed, and that it cannot reach the speed at which lift-propelled gaits still cruise.

A complementary check replays the 27 feasible optima of the drag-only model with lift switched on. For 21 of them the lift thrust share exceeds 0.1 (median 1.26), so the same leg motions are mainly lift-propelled once lift exists, and 26 no longer stay straight and level. Whether a stroke is drag-based or lift-based is therefore a property of the physics as much as of the motion: a flipper that crosses the water at an angle produces lift whether the model includes it or not. Paddling strokes found in a drag-only model cannot be transferred unchanged to a real flipper.

5.5 Summary

Within this model the answer to RQ1 is clear in direction. When lift is part of the physics and not suppressed, the optimizer selects lift-propelled strokes in all 101 feasible optima (22 in this chapter, the others in the coordination study of Appendix B). At equal speed they need one half to one third of the power of the best strokes of a drag-only model. When drag-based strokes are forced within the lift model, the few that are feasible (3 of 30 searches) need four to ten times the power of the lift-propelled strokes and do not reach 0.30 m s^{-1} ; in this model a drag-based stroke therefore costs at least about twice and up to ten times as much. Whether a stroke

is drag- or lift-based depends on the physics: most paddling strokes of the drag-only model become lift-propelled when lift is added. Which coordination of the legs is best is a separate question, which depends mainly on the attitude limits; it is reported in Appendix B. Three caveats limit these statements to the model: the coefficients and the lift slope are not calibrated, the flat-plate lift has no aspect-ratio correction although the flipper has $AR \approx 1.4$, and the searches are not fully converged, particularly the constrained search for drag-based strokes. Chapters 6 and 8 therefore test the central claim, that lift-based flapping beats drag-based paddling, with resolved flow simulations of the real leg.

Chapter 6

Flow Simulation Methods

Two flow solvers are used. Single-leg simulations with OpenFOAM resolve the flow around one propulsor and provide the thrust and power data of Chapter 8. Whole-robot simulations with the real-time lattice-Boltzmann solver Flare Live provide the reference fluke motion and qualitative trends of the free-swimming robot (Appendix C). This chapter describes both.

6.1 Single-Leg CFD with Overset Grids

6.1.1 Governing Equations and Solver

The flow of water ($\rho = 1000 \text{ kg m}^{-3}$, $\nu = 1.0 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$) is modelled as incompressible, laminar and single-phase. The Reynolds numbers of 10^3 – 10^4 (Section 3.1) lie in the laminar-to-transitional range where the forces are dominated by the separated vortices at the edges, and no turbulence model is used. The incompressible Navier–Stokes equations are solved with `overPimpleDyMFoam` of OpenFOAM v2606 [Wel+98], a finite-volume solver with the PIMPLE pressure–velocity coupling and overset (Chimera) grids [Win+20]. The time step is adapted to a maximum Courant number of 0.8.

Overset grids are chosen because the propulsors rotate by up to 120° relative to the hull per cycle. A body-fitted component mesh moves rigidly with the propulsor inside a stationary background mesh. In every time step, background cells covered by the propulsor are removed (hole cutting), and the solutions are coupled by interpolation at the fringe cells. The inverse-distance stencil is used; for some cases it crashed in the stencil update, and those cases were continued from their last saved time with the tracking inverse-distance variant. As a guard against failed hole cutting, the number of hole cells is checked in the first time step and the case is restarted with another stencil if it exceeds 3 % of all cells or is below 50.

6.1.2 Domain, Boundary Conditions and Motion

The simulations are carried out in a frame that moves with the hull. The forward speed U enters as a uniform inflow at the upstream boundary; the downstream boundary has a fixed pressure and an inlet–outlet condition for the velocity. The top of the domain is a slip wall 12 cm above the fluke, a rigid lid that represents deep water without a free surface; the lateral and bottom

Table 6.1: Mesh parameters of the single-leg simulations. Sizes are the cell edge lengths in the refined region around the propulsor.

	Fluke (L cases)	Fan paddle (D cases)	Validation foil
Component mesh (surface)	1.2 mm (0.6 mm)	2.5 mm (1.25 mm)	2.5 mm
Background mesh	2.5 mm	4 mm	5 mm
Surface cells per length	≈ 50 per chord	≈ 34 per radius	≈ 40 per chord
Cells, component + backgr.	$1.8 \times 10^5 + 5.4 \times 10^5$	–	–
Cycles simulated / averaged	3 / 2	3 / 1	3 / 2
Start-up	0.5-cycle smooth step	none	none

boundaries are also slip walls far from the propulsor. The propulsor's rigid-body motion is prescribed as a table of positions and orientations from the leg kinematics of Section 4.3. To avoid pressure spikes and collapsing time steps from an impulsive start, the fluke motion starts from rest with a smooth step over the first half cycle. Three cycles are simulated; means are taken over the last two for the fluke and over the third for the fan paddle.

Only the propulsor is simulated: the fluke without its foot rod, and the fan paddle with its thin 3 mm rod but without the crank and parallelogram. The linkage contributes only drag, which is assumed to be the same for both propulsors and is accounted for in the resistance of the self-propulsion balance (Section 8.5). The rod contributes about 1 % of the paddle thrust. The hull resistance used in Section 8.5 comes from separate two-phase simulations of the bare hull with a free surface, which are not described further here.

6.1.3 Meshes

Both meshes are built with snappyHexMesh from the propulsor surface. For the fluke, a clean CFD geometry is used: the pin hole and the end-stop arm of the printed part are removed and the 0.1 mm trailing edge is truncated at 95 % chord, because these details produced degenerate cells that reduced the time step below 10^{-4} s. The resulting blade deviates from the Flare geometry by a median of 0.12 mm. The mesh parameters are listed in Table 6.1.

The fan-paddle mesh is coarser than the fluke mesh. It was set up earlier, when the paddle strokes were designed, and was kept to preserve comparability across the paddle design iterations. Its consequence for the comparison is discussed in Section 7.4.

6.1.4 Two-State Composite for the Folding Fan

Simulating the folding of the fan would require a deforming geometry. Instead, each paddle stroke is simulated twice with the same motion: once with the fan open and once with the fan closed. The cycle force history is then composed from the two runs,

$$\mathbf{F}(\tau) = \begin{cases} \mathbf{F}_{\text{open}}(\tau), & \tau_o \leq \tau < \tau_c, \\ \mathbf{F}_{\text{closed}}(\tau), & \text{otherwise,} \end{cases} \quad (6.1)$$

where τ is the cycle phase and τ_o, τ_c are the opening and closing phases (Figure 6.1). The opening and closing are thus instantaneous, and each run keeps its own wake. The composite is a first-order model of the folding. Its advantage is that the fold timing can be evaluated after the simulation for any pair (τ_o, τ_c) at no extra cost. It has two systematic errors, both of which make it optimistic.

Neglected change of added mass. When the fan opens, the water around it must be set in motion; when it closes, that water is released. In (6.1) the open run starts with a flow that has already been established while the fan was open, and the closed run ends without having to release anything. With the water-frame velocity V of the fan (positive forward) and the difference Δm_a between the added masses of the open and the closed fan, the missing impulse on the paddle per cycle lies between two limits:

$$\Delta J = \begin{cases} \Delta m_a (V_c - V_o) & \text{potential flow: the released momentum returns to the paddle,} \\ -\Delta m_a V_o & \text{full shedding: the released momentum stays in the wake,} \end{cases} \quad (6.2)$$

where V_o and V_c are the velocities at opening and closing. In a separated flow, part of the released momentum is shed as a stopping vortex, so the true value lies between the two limits; the opening cost $-\Delta m_a V_o$ has no such escape and is present in both. The corrected mean thrust is reported as the range $\bar{T} + \Delta J/T$ over both limits. The added mass of the open fan was identified from the acceleration phase of the open-paddle simulation at $U = 0$ by fitting $F_x = -m_a \dot{v}_x - \frac{1}{2} \rho C_N S v_x |v_x|$ to the force history, which gives $m_a = 38 - 49$ g depending on the fitting window and the reference point, close to the 38 g of a disc of the same area. With about 5 g for the fan folded to 12 mm, $\Delta m_a = 38$ g is used for V3trap and 20 g for the robot gait, whose fan folds only to 30 mm; the uncertainty of Δm_a is about $\pm 15\%$. When the fan switches while it is at rest in the hull frame, the correction does no work in that frame, and the power $\bar{P}$ is not corrected. V3trap closes at $\tau_c = 0.33$ while the fan still moves; the corresponding work, of order $\frac{1}{2} \Delta m_a V_c^2$ per cycle, is not included in $\bar{P}$.

Two consequences follow. First, switching at the reversal points of the hip motion, where V equals the forward speed U , is free of correction at $U = 0$ but costs $\Delta m_a U$ per cycle at speed. Second, opening the fan before the reversal, during the deceleration of the recovery, credits the paddle with the momentum that the open fan releases while it decelerates, without charging it for having set that water in motion. The composite therefore rewards early opening artificially, and all paddle results in this thesis use $\tau_o = 0$.

Wake of the open run. During the power stroke the open-paddle run moves through the wake of its own open recovery, which has pushed water forward; the real paddle recovers folded and leaves a much weaker wake. This raises the relative velocity of the power stroke in the composite. The effect is not quantified and is listed as a limitation.

6.1.5 Post-Processing

Pressure and viscous forces are integrated over the propulsor surface in every time step, and moments are taken about the hip axis O_2 . The thrust, the input power and the efficiency follow from (3.9)–(3.11), with the velocity of the fluke's pin or the paddle's reference point and the angular velocity taken from the motion table. The pitch moment about the fluke pin gives the hinge moment required from a pitch actuator or spring. The angle of attack is evaluated from (3.5) with the pin velocity and the pitch angle and is reported either as the 75th percentile of $|\alpha|$ over the cycle, α_{75} , or as the median over the middle part of each half stroke, away from the stroke reversals. For the L1 case with $\alpha = 10^\circ$, whose hinge path had velocity kinks at the seven control points of the trajectory, the resulting added-mass spikes in the force history were filtered before averaging; this case is marked as less certain in the results.

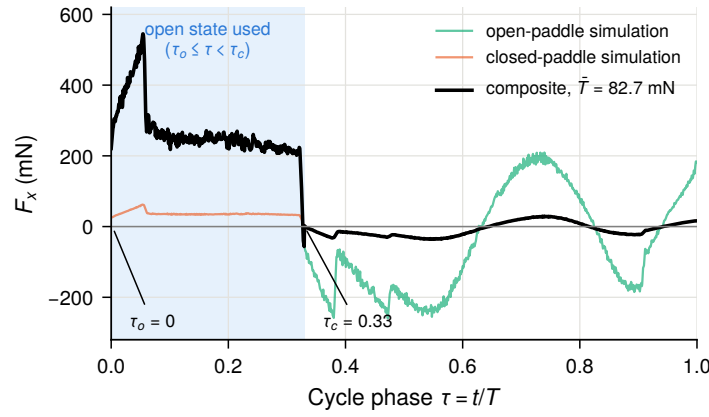


Figure 6.1: Two-state composite of the fan paddle for V3trap at $U = 0$. The open-paddle simulation is used between $\tau_o = 0$ and $\tau_c = 0.33$ (shaded), the closed-paddle simulation for the rest of the cycle. The composite has a cycle-mean thrust of 82.7 mN before the added-mass correction of (6.2).

Table 6.2: Single-leg simulation cases.

Case	Description	U (m s^{-1})
L0	fluke, reference motion from Flare Live	0.223
L2	fluke, L0 motion unchanged (fixed pitch schedule)	0, 0.08, 0.15, 0.30
L1	fluke, angle-of-attack law $\alpha = 10^\circ, 28^\circ / 20^\circ$	0.223 / 0.40
L3	fluke, speed-scheduled pitch $\theta_0 = \gamma_{\max}/2$	0.30, 0.40
L0c	L0 on a coarser mesh (all cell sizes $\times 1.33$)	0.223
Robot gait	fan paddle, current stroke (30 mm folded width)	0, 0.08, 0.15
V3trap	fan paddle, optimized stroke (12 mm folded width), $\tau_o = 0$, $\tau_c = 0.33$	0, 0.08, 0.15, 0.223, 0.30
V	paddle design iterations V1, V2, V3, V3c12, V3c06 (Section 8.1)	0
Val	validation: NACA 0012 heaving and pitching foil (Section 7.1)	0.40

6.1.6 Case Matrix

Table 6.2 lists all single-leg simulations. Each drag-type (D) entry consists of an open and a closed run.

The fluke cases took 3–4 h each and every paddle state about 2.5 h on eight cores; the validation case, with a foil of 0.6 m span, took about 43 h.

6.2 Whole-Robot Simulation in Flare Live

Flare Live [Man26] is a commercial solver that couples a lattice-Boltzmann flow solver on adaptive block grids with a multibody model of the robot and runs close to real time on a GPU. The robot is imported as rigid bodies connected by joints; each joint is driven by a position controller that follows the commanded trajectory. The legs and flukes are immersed in the fluid. The hull is not resolved in the fluid; its resistance is applied as an external force $D = K U|U|$. Three values of K are used: $3.95 \text{ N s}^2 \text{ m}^{-2}$ from the original Flare model, 0.65 corresponding to the hull-and-linkage estimate D_{mid} of Section 8.5 and 0.27 corresponding to the bare hull. The

model mass is 863 g, about twice the measured mass of the hardware (418.8 g); the difference was not corrected because Flare Live is used only for trends. The lattice spacing is 3 mm, with a 2 mm lattice as a sensitivity test.

The legs move in a trot at 2 Hz (2–3 Hz in a frequency sweep). The fluke pitch is actively controlled by an angle-of-attack law: from the measured heave velocity of the pin and the current swimming speed, the pitch is set so that (3.5) gives a target α of 10° , 18° or 28° , within the fluke's end stops of -75° and 60° .

Two kinds of runs are made:

- *Self-propelled runs.* The robot swims freely from rest; speed, power and attitude are averaged over the steady part. Power is the mechanical power of the joint drives, and the cost of transport is $P_j/(mgU)$ with the joint-drive power P_j .
- *Tow runs.* The robot is held fixed in a uniform stream of speed U with one leg flapping (the L0 motion) and the others idle. The fluid force on the fluke is obtained from the joint reaction at the pin, transformed into the fluke frame, and averaged over $t = 3 - 9$ s. A run with the fluke idle gives the reference drag.

The tow runs use exactly the motion of the CFD case L0 and its speed series, so that the fluke force of the two solvers can be compared directly (Section 7.5).

Chapter 7

Verification and Validation

The single-leg CFD provides all quantitative statements of Chapter 8. This chapter checks it in three ways: against a flapping-foil experiment (validation), on a coarser mesh (spatial discretization) and from cycle to cycle (temporal convergence). It closes with the consequences for the drag-type cases, which were not part of the mesh study, and with a check of the real-time solver Flare Live against the CFD.

7.1 Validation with a Heaving and Pitching Foil

Case. Schouveiler et al. [SHT05] measured the thrust and efficiency of a NACA 0012 foil in combined heave and pitch in a towing tank, with error bars. One of their points is reproduced with the same solver, the same overset method and comparable resolution as the fluke cases: chord $c = 0.1$ m, span 0.6 m with free tips, heave amplitude $h_0/c = 0.75$, pitch axis at $c/3$, pitch leading heave by 90° , $St = 0.25$, maximum angle of attack 15° , $U = 0.4 \text{ m s}^{-1}$ ($f = 0.667 \text{ Hz}$) and $Re = 4 \times 10^4$. The component mesh has 2.5 mm cells ($c/40$), the background 5 mm. Three cycles were simulated without a start-up ramp.

Results. The streamwise force settles after the first cycle (Figure 7.1a); cycles 2 and 3 differ by less than 0.1 %. The simulated motion reaches $\alpha_{75} = 14.6^\circ$, consistent with the prescribed 15° , and the mean lateral force is 1.3 % of the thrust. The mean thrust is 1415 mN and the input power 946 mW. Table 7.1 compares the coefficients with the experiment.

The thrust coefficient is within 8 % of the measurement, inside the ± 10 –15 % usually reported for flapping-foil CFD, and is taken as the main criterion. The efficiency is 18 % too low. Three causes act in the direction observed. (i) The simulated foil has free tips and an aspect ratio of 6, while the experiment was close to two-dimensional; a finite-wing estimate $AR/(AR + 2) = 0.75$ suggests tip losses of 10–20 %, which affect efficiency more than thrust because tip vortices cost power. (ii) At $Re = 4 \times 10^4$ the boundary layer is transitional; a laminar model separates earlier and overestimates the power. (iii) The mesh ($c/40$) is coarser than in the fluke cases ($c/50$). Because this thesis compares propulsors simulated with the same solver and settings, such systematic deviations affect both propulsors in the same direction to first order; the tip losses

Table 7.1: Validation against the experiment of Schouveiler et al. [SHT05] (NACA0012, $St = 0.25$, $\alpha_{\max} = 15^\circ$).

	CFD (this work)	Experiment	Deviation
Thrust coefficient C_T	0.295	0.32 ± 0.01	−8 %
Froude efficiency η	0.60	0.73 ± 0.04	−18 %

and the laminar separation act mainly on the lift-based fluke, so its advantage is, if anything, underestimated. Absolute efficiencies should be read as conservative.

7.2 Mesh Sensitivity

The reference fluke case L0 was repeated on a mesh in which all cell sizes were multiplied by $r = 1.33$ (component 1.6 mm, background 3.3 mm). The coarse run was stopped at $t = 0.91$ s, so both runs are compared over the common window $t = 0.41 - 0.91$ s. The force histories are nearly identical in shape (Figure 7.1c). The coarse mesh gives 7.3 % less thrust (19.75 versus 21.30 mN), 3.9 % less power and 3.4 % lower efficiency.

With two meshes the order of convergence cannot be computed. Assuming second order ($p = 2$) and a safety factor of $F_s = 1.25$ [Roa94], the grid convergence index of the thrust on the fine mesh is

$$GCI = F_s \frac{|\varepsilon|}{r^p - 1} = 1.25 \times \frac{0.073}{1.33^2 - 1} \approx 12\%. \quad (7.1)$$

The discretization uncertainty of the fluke thrust is therefore taken as about $\pm 12\%$, and that of power and efficiency as about $\pm 6\%$. Since the coarse mesh gives lower thrust, the fine-mesh values are more likely to be slightly low than high.

7.3 Temporal Convergence

For every fluke case, the mean thrust of the last cycle differs from that of the second-last cycle by at most 0.61 % (L2 at $U = 0$), and by at most 0.31 % for all other cases (Figure 7.1d). The validation case differs by less than 0.1 %. Averaging over the last two cycles after a half-cycle start-up is therefore sufficient; the periodic state is reached within the first full cycle. The fan-paddle cases were evaluated in their third cycle. For the robot gait at 0.08 m s^{-1} , cycles 2 and 3 give 7.3 and 5.2 mN; this case, whose thrust is small, has a correspondingly larger uncertainty.

7.4 Consequences for the Drag-Type Cases

No separate mesh study was run for the fan paddle, and its mesh is coarser than that of the fluke (Table 6.1). Three arguments bound the effect on the conclusions.

1. *Flow physics.* The paddle is a thin plate with sharp edges moved broadside. Its separation lines are fixed at the edges, whereas the fluke's thrust depends on a leading-edge vortex and leading-edge suction that are more sensitive to resolution. The fluke's $\pm 12\%$ is a plausible upper bound for the paddle.
2. *Low speed.* Up to 0.15 m s^{-1} the optimized paddle produces 1.6–2.0 times the thrust of the fluke. An error of $\pm 12\%$ does not change this ranking.

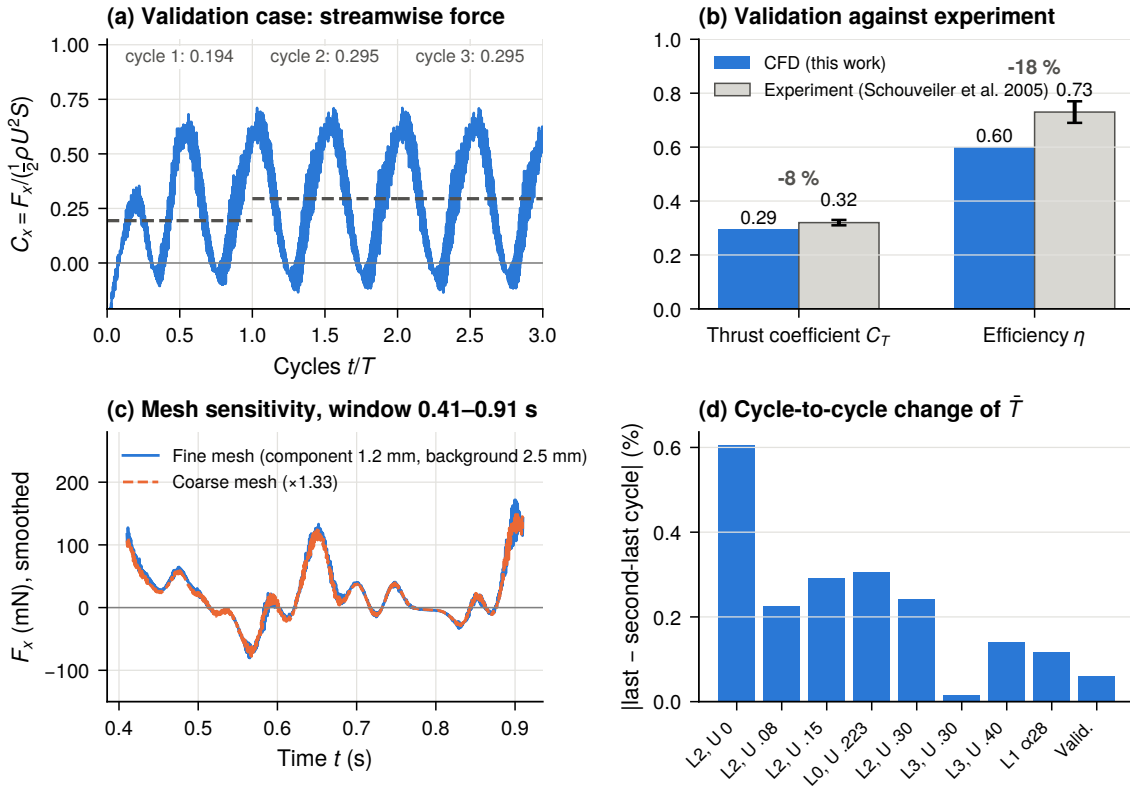


Figure 7.1: Verification and validation of the single-leg CFD. (a) Validation case: streamwise force coefficient with cycle means (dashed). (b) Thrust coefficient and efficiency against the experiment of Schouveiler et al. [SHT05]. (c) Fluke L0 on the fine and the coarse mesh over the common time window. (d) Change of the mean thrust between the last and the second-last cycle for all fluke cases and the validation case.

3. *Crossover.* Near the crossover the thrust falls with slopes of about -110 (fluke) and -340 mN s^{-1} (paddle). A $\pm 12\%$ error of the paddle thrust (about 2.5 mN) shifts the crossover by about 0.01 m s^{-1} ; opposite errors of $\pm 12\%$ on both sides shift it by about 0.02 m s^{-1} . The uncertainty of the added-mass correction (Section 6.1.4) adds less than 0.01 m s^{-1} .

The coarser paddle mesh is nevertheless a limitation and is listed in Section 9.4.

7.5 Cross-Check of the Real-Time Solver

Flare Live (Section 6.2) simulates the whole robot in real time but on a lattice of $2-3 \text{ mm}$. Before any of its results can be used, its fluke force is checked against the CFD. In the tow runs, the fluke performs exactly the L0 motion of the CFD. Its angle of attack agrees ($\alpha_{75} = 19.7^\circ$ versus 19.9° at 0.223 m s^{-1}), but its thrust does not (Figure 7.2). On the 3 mm lattice, Flare Live gives 20.0 , 8.5 and -0.1 mN at 0 m s^{-1} , 0.1 m s^{-1} and 0.223 m s^{-1} , against 45.5 , about 34 (interpolated) and 21.4 mN in CFD (Table C.1): 44% , 25% and practically zero. The deficit is about $21-26 \text{ mN}$ at all three speeds, which is the order of the lift-based thrust itself.

The idle fluke, held at its zero position, has a drag of 12.6 mN at 0.223 m s^{-1} . At its incidence of -25° this corresponds to a drag coefficient of about 0.45 based on the planform, a normal value for a stalled plate. The thrust increment of flapping over idling is 12.5 mN , still well

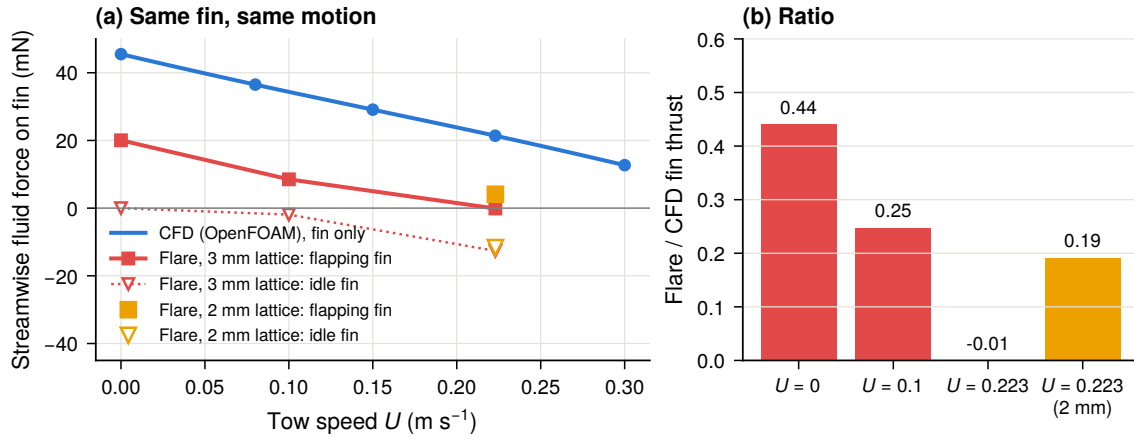


Figure 7.2: Fluke force in Flare Live and in CFD for the same fluke and motion. (a) Streamwise force versus tow speed, flapping and idle. (b) Ratio of the Flare to the CFD thrust.

below the CFD thrust.

On a 2 mm lattice the fluke thrust at 0.223 m s^{-1} rises to 4.1 mN (19 % of CFD) and the idle drag hardly changes (11.6 mN). The result is not converged, and a Richardson extrapolation from the two lattices gives only about 8–12 mN, which indicates that the lattices are far from the asymptotic range. The cause is resolution: the 34 mm chord spans only 11 cells at 3 mm and 17 at 2 mm, while the boundary layer at $Re \approx 7600$ is about 0.4 mm thick. The leading edge, where the suction that drives lift-based thrust acts, is not resolved. Resolving it would need a lattice of 1 mm or finer at many times the cost, which defeats the purpose of a real-time solver.

Flare Live is therefore used only for the reference motion of the fluke and for qualitative trends of the free-swimming robot, which are collected in Appendix C; all quantitative statements rest on the CFD.

7.6 Summary

Table 7.2 collects the checks of this chapter.

Table 7.2: Summary of the verification and validation.

Check	Result
Validation (foil experiment)	C_T -8% , η -18% ; deviations mainly from free tips and laminar model
Mesh sensitivity (fluke)	$\bar{T}$ -7.3% on the $\times 1.33$ coarser mesh, $GCI \approx \pm 12\%$
Temporal convergence	fluke: last two cycles differ by $\leq 0.61\%$; robot gait at 0.08 m s^{-1} : 7.3 vs. 5.2 mN
Drag-type mesh	not studied; ranking insensitive, crossover within $\pm 0.02 \text{ m s}^{-1}$
Real-time solver (Flare Live)	fluke thrust 0–44 % of CFD on 2–3 mm lattices; used for trends only

The single-leg CFD reproduces the thrust of a flapping foil well and underestimates its efficiency. Its numerical uncertainty is dominated by spatial discretization, about $\pm 12\%$ in thrust. Differences between propulsors of less than about 15 % are therefore not interpreted as significant in Chapter 8.

Chapter 8

Results

This chapter reports the single-leg CFD results and what follows from them for the whole robot. It first develops the drag-type stroke (Section 8.1) and characterizes the fluke (Section 8.2), then compares both across speed (Section 8.3), improves the fluke by scheduling its pitch with speed (Section 8.4), predicts the cruising speed of the robot (Section 8.5) and compares it with a first hardware test (Section 8.6). All forces and powers are per leg and for the propulsor alone unless stated otherwise.

8.1 Design of the Drag-Type Stroke

The drag-type stroke was optimized at $U = 0$ one factor at a time. The quasi-steady model of Section 3.2 served as a surrogate that proposed each change, and CFD evaluated it; the fold timing was then optimized over both switching phases on the two-state composite. Table 8.1 lists the steps.

Direction of the stroke. Centring the sweep on the vertical makes the paddle move nearly straight backwards with its face normal to the motion. From V1 to V3 the inclination of the power-stroke force from the swimming direction fell from 19.5° to 7.9° , and the ratio of mean vertical to mean horizontal force from 0.52 to 0.28. V2 shows that shortening the power stroke alone loses thrust when the recovery cannot become correspondingly cheaper.

Folded width. Folding the fan to 12 mm instead of 30 mm reduced the recovery drag from -10.7 to -3.2 mN and raised the mean thrust by 15 %. The effective area ratio a , defined as the ratio of the mean force of the closed-only to the open-only run, fell from 0.48 to 0.11. With $a = 0.11$, (3.3) gives $x^* = 2.2$ and $\text{DUTY}^* = 0.69$: the recovery should be much faster than the power stroke, which the servo speed does not allow. The DUTY of the final stroke, 0.382, was therefore not taken from (3.3) but found in the CFD optimization within the joint speeds that the servos allow (Table A.3). Feathering the folded fan to a frontal width of 5.6 mm added only 3 %: the recovery drag is proportional to the width, but at 12 mm it is only about 6 % of the mean thrust. A feathering mechanism is therefore not worthwhile.

Table 8.1: Design steps of the drag-type stroke at $U = 0$. Each step changes one aspect of the previous one. $\bar{T}$ is the cycle-mean thrust of the two-state composite. All steps before the last switch the fan at the reversal points of the hip motion, where the added-mass correction (6.2) vanishes at $U = 0$; for the last step the range of the correction is given.

Stroke	Change	T (s)	DUTY	$\bar{T}$ (mN)	$\bar{P}$ (mW)
Robot gait	baseline, 30 mm folded, 60° sinusoidal sweep	1.25	0.50	23.4	6.7
V1	shorter period, larger thrust	0.89	0.49	51.2	–
V2	shorter power stroke	–	0.35	34.2	–
V3	sweep centred on the vertical (120–70°), slow curl	0.81	0.45	49.3	16.2
V3c12	fan folded to 12 mm instead of 30 mm	0.81	0.45	56.7	13.7
V3c06	folded and feathered, 5.6 mm frontal width	0.81	0.45	58.6	13.1
V3trap	V3c12 with trapezoidal velocity law (45 ms ramps)	0.72	0.382	75.0	20.8
V3trap, timed	fan closed at the start of the deceleration, $\tau_c = 0.33$	0.72	0.382	73–83	21.6

Velocity law. Replacing the cosine hip velocity by a trapezoidal law with 45 ms ramps at the same peak rate and sweep increased the power-stroke impulse by 18 %, as predicted by the quasi-steady $\int v^2 dt$, and the mean thrust by 32 %, because the period also shortened from 0.81 to 0.72 s. The price is a higher peak hip moment (35.6 instead of 19.3 mN m, $\times 1.84$), a higher peak force and 52 % more power; the thrust per power fell from 4.1 to 3.6 NW^{-1} . The effective normal-force coefficient of the power stroke in the CFD is 2–3 times the quasi-steady value and is highest at the start of the stroke, as expected for a starting plate [KG11]; it is the same for both velocity laws.

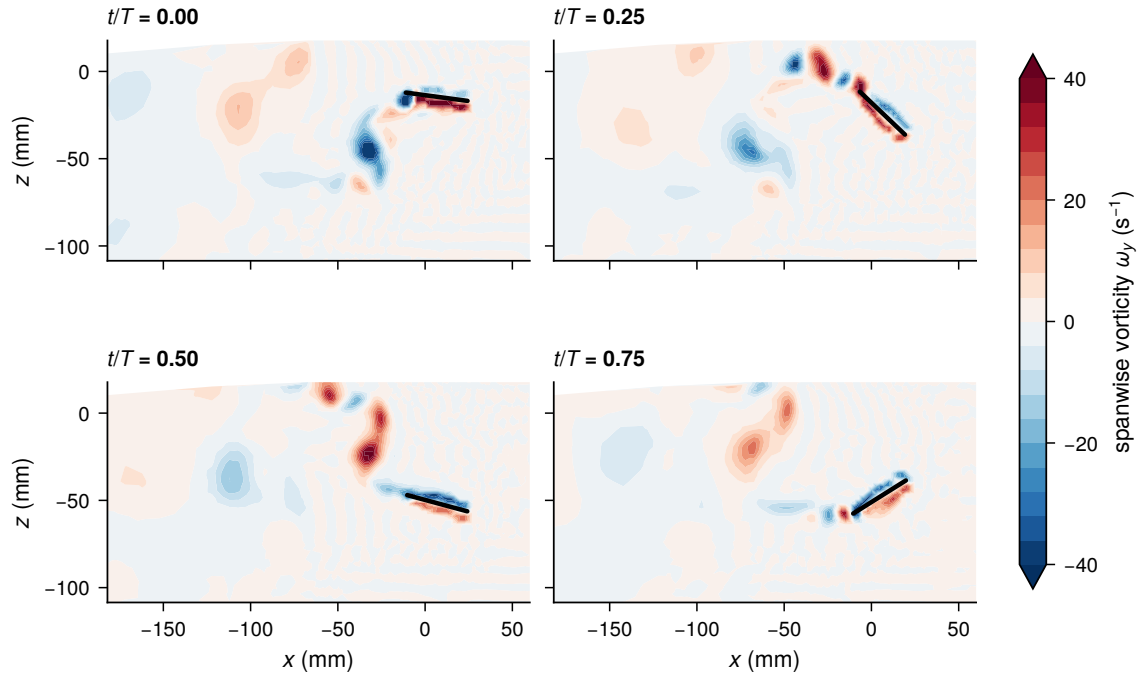
Fold timing. The two-state composite (Section 6.1.4) allows the fold timing to be scanned after the simulation. Without correction, the best pair is $\tau_c = 0.33$, $\tau_o = -0.06$ with 87.2 mN. The gain from opening before the reversal, however, is the artefact described in Section 6.1.4, and the fan is therefore opened at the reversal, $\tau_o = 0$. Closing the fan early, at the start of the power-stroke deceleration ($\tau_c = 0.33$, 0.86 DUTY), raises the uncorrected composite from 75.0 to 82.7 mN: the decelerating open fan is no longer pulled backwards by the water it has set in motion. Whether that water keeps its momentum depends on how much of it is shed as a stopping vortex, so the added-mass correction (6.2) gives a range of 73–83 mN. At rest, early closing is therefore worth between -2 and $+8$ mN. At forward speed it is never worse and up to 10 mN better, because after $\tau \approx 0.33$ the paddle moves backwards more slowly than the oncoming flow and an open fan would only produce drag; at 0.223 m s^{-1} , closing at $\tau_c = 0.33$ gives 24–26 mN, against 14–26 mN for closing at the reversal. The final stroke, called V3trap in the rest of this thesis (Figures 4.3 and 6.1a), produces 73–83 mN per leg at rest, 3.1–3.5 times the robot’s current gait, at 21.6 mW and 3.4–3.8 NW^{-1} . These rules agree with those found for crustacean and webbed-foot propulsors: open fully at the start of the power stroke, close early, keep transitions short [San+26; LGB24].

8.2 The Lift-Type Fluke with a Fixed Pitch Schedule

Table 8.2 lists the fluke results with the L0 motion at five speeds. At rest the inflow is purely vertical, the fluke is stalled ($\alpha_{75} = 73^\circ$) and acts as a small paddle; it produces 45.5 mN at

Table 8.2: Fluke with the L0 motion (fixed pitch schedule) at different speeds. St is based on the peak-to-peak pin excursion of 39.2 mm.

U (m s^{-1})	0	0.08	0.15	0.223	0.30
$\bar{T}$ (mN)	45.5	36.5	29.1	21.4	12.7
$\bar{P}$ (mW)	18.1	17.1	16.6	16.3	15.6
η	–	0.171	0.263	0.293	0.245
$\bar{T}/\bar{P}$ (N W^{-1})	2.51	2.13	1.75	1.31	0.81
St	–	0.98	0.52	0.35	0.26

**Figure 8.1:** Spanwise vorticity in the mid-span plane of the fluke (L0, $U = 0.223 \text{ m s}^{-1}$) at four phases of the third cycle, hull frame; the flow is from right to left relative to the fluke. Black line: fluke chord.

18.1 mW, about twice the thrust of the robot gait with the fan paddle. With increasing speed the inflow angle and the angle of attack fall, and the thrust decreases almost linearly, to 12.7 mN at 0.30 m s^{-1} . The input power stays between 15.6 and 18.1 mW. The efficiency rises to a maximum of 0.29 at 0.223 m s^{-1} , the speed for which L0 was designed ($St \approx 0.35$ based on the pin excursion, $\alpha_{75} = 19.9^\circ$). At 0.30 m s^{-1} the mid-stroke angle of attack drops to about 10° , the efficiency falls to 0.245, and a mean vertical force of -29 mN appears, more than twice the thrust.

The wake at 0.223 m s^{-1} (Figure 8.1) shows the expected pattern of a thrust-producing foil: each half stroke sheds a vortex of alternating sign from the trailing edge, and the vortices form a reverse von Kármán street that convects downstream.

Additional pitch schedules for target angles of attack of 10° and 28° at 0.223 m s^{-1} produce more thrust than L0 but at lower efficiency (0.20 and 0.24, Table A.2); among the cases at this speed the efficiency is highest for $\alpha_{75} \approx 20^\circ$, in line with the efficient range of flapping foils [RHT03].

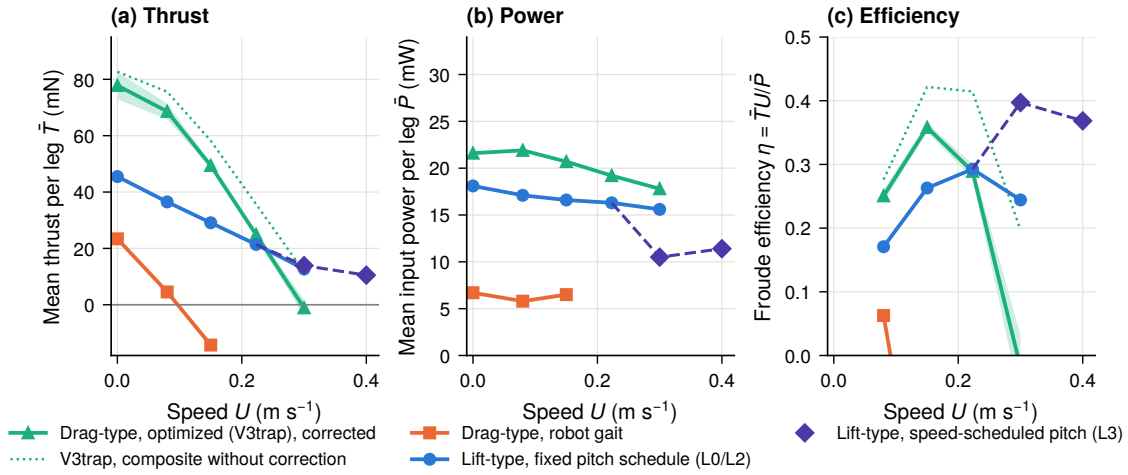


Figure 8.2: Single-leg (a) thrust, (b) input power and (c) Froude efficiency of the propulsors versus speed. Shaded: range of the added-mass correction of the paddle composites (6.2); dotted: V3trap composite without correction.

Table 8.3: Single-leg thrust $\bar{T}$ (mN) / input power $\bar{P}$ (mW) / Froude efficiency η across speed. Paddle values include the added-mass correction (6.2) as a range; the uncorrected composite of V3trap gives 82.7, 75.6, 58.2, 35.7 and 11.7 mN.

U (m s^{-1})	Paddle, robot gait	Paddle, V3trap	Fluke, fixed (L0/L2)	Fluke, scheduled (L3)
0	23.4 / 6.7 / –	73–83 / 21.6 / –	45.5 / 18.1 / –	–
0.08	4–5 / 5.8 / 0.05–0.07	66–71 / 21.9 / 0.24–0.26	36.5 / 17.1 / 0.17	–
0.15	< 0 / 6.5 / –	48–50 / 20.7 / 0.35–0.37	29.1 / 16.6 / 0.26	–
0.223	–	24–26 / 19.2 / 0.28–0.30	21.4 / 16.3 / 0.29	–
0.30	–	–4–2 / 17.8 / ≈ 0	12.7 / 15.6 / 0.24	13.9 / 10.5 / 0.40
0.40	–	–	–	10.5 / 11.4 / 0.37

8.3 Drag versus Lift across Speed

Figure 8.2 compares thrust, power and efficiency of the propulsors from rest to 0.40 m s^{-1} ; Table 8.3 gives the values.

Low speed. Up to about 0.15 m s^{-1} the optimized paddle is clearly superior. It produces 1.6–2.0 times the thrust of the fluke at 19–28 % more power, and at 0.15 m s^{-1} it reaches an efficiency of 0.35–0.37, higher than the fluke’s maximum of 0.29. Per area there is no difference at rest: the paddle produces $3.9\text{--}4.5$ and the fluke 4.1 mN cm^{-2} . At 0.223 m s^{-1} the fluke already produces more per area (1.9 versus $1.3\text{--}1.4 \text{ mN cm}^{-2}$). The robot gait, in contrast, loses its thrust almost immediately: at 0.08 m s^{-1} only $4\text{--}5 \text{ mN}$ remain, and at 0.15 m s^{-1} the net force is a drag.

Speed limit of the paddle. The paddle’s thrust falls steeply with speed and vanishes at about 0.30 m s^{-1} . The split into power stroke and recovery (Figure 8.3, values in Table A.1) shows why. The power-stroke thrust of V3trap decreases by 42 %, from 88.0 to 50.9 mN , because the relative velocity during the power stroke falls. The recovery drag, however, grows more than sevenfold, from -5.3 to -39.1 mN , because the folded paddle and the rod now move forward against the oncoming flow. At speed, opening the fan also costs the momentum $\Delta m_a U$ per

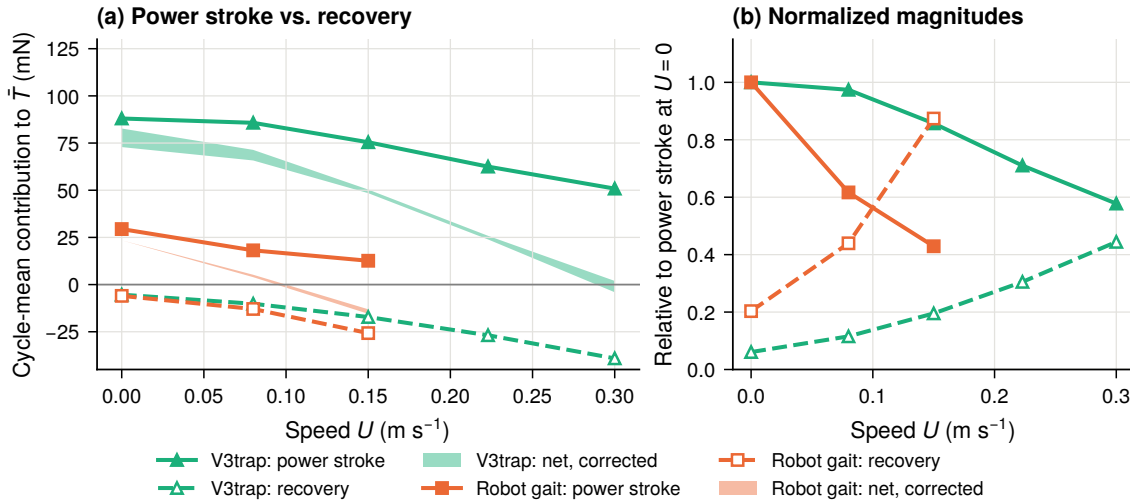


Figure 8.3: Contributions of power stroke and recovery to the mean thrust of the fan paddle. (a) Absolute values for V3trap and the robot gait; shaded: net thrust including the added-mass correction (6.2). (b) Magnitudes relative to the power-stroke thrust at $U = 0$.

cycle, 12–16 mN at 0.223–0.30 m s⁻¹. At 0.30 m s⁻¹ the recovery and the opening together consume the whole power-stroke thrust. For the robot gait, with its wider folded fan and slower stroke, the recovery drag exceeds the power-stroke thrust already at about 0.1 m s⁻¹. The speed limit of a drag-type paddle on this leg is therefore set by what the paddle has to do outside the power stroke, not by the speed of the paddle in the power stroke.

Crossover. The fluke loses thrust more slowly, because it neither recovers against the flow nor changes its shape: its thrust at 0.30 m s⁻¹ is 28 % of its value at rest, whereas the paddle’s is close to zero. Interpolating the curves, the thrust of the fluke exceeds that of V3trap above 0.23–0.24 m s⁻¹, and its efficiency above 0.22–0.23 m s⁻¹. This holds for both pitch schedules, because the speed-scheduled pitch was simulated only at 0.30 m s⁻¹ and 0.40 m s⁻¹ and shares the L0 data below. Including the discretization uncertainty of Section 7.4, the crossover lies at 0.24 ± 0.02 m s⁻¹ for thrust and at 0.22 ± 0.02 m s⁻¹ for efficiency. Without the added-mass correction, the crossovers would lie at 0.295 and 0.285 m s⁻¹; the correction moves them by about 0.06 m s⁻¹ but does not change their existence.

8.4 Speed-Scheduled Fluke Pitch

With the fixed L0 motion, the fluke’s angle of attack falls with speed, which explains its efficiency drop above 0.223 m s⁻¹. The speed-scheduled pitch (3.8) restores the angle of attack (Figure 8.4). At 0.30 m s⁻¹ it reduces the pitch amplitude from about 40° to 21.6°, and the angle of attack rises accordingly: α_{75} increases from 13.7° to 21.0° (Tables 8.4 and A.2).

At 0.30 m s⁻¹ the scheduled pitch raises the thrust by 9 %, lowers the power by 33 % and raises the efficiency from 0.245 to 0.40, while the paddle produces almost no thrust at this speed. It also removes the mean vertical force (−29 to −1 mN), reduces the peak hinge moment by 40 % and flattens the force history: the peak-to-peak streamwise force falls from about 600 to 190 mN. At 0.40 m s⁻¹ the efficiency is 0.37. Over the three designed speeds the fluke efficiency is 0.29 (L0, 0.223), 0.40 (L3, 0.30) and 0.37 (L3, 0.40), with a maximum near 0.30 m s⁻¹. At this speed the Strouhal number based on the pin excursion is 0.26, inside the

Table 8.4: Speed-scheduled pitch (L3) compared with the fixed L0 motion (L2). $\bar{F}_z$: mean vertical force; $|M_h|$: peak hinge moment about the pin.

Case	U (m s ⁻¹)	$\bar{T}$ (mN)	$\bar{P}$ (mW)	η	$\bar{F}_z$ (mN)	$ M_h $ (mN m)
L2, fixed	0.30	12.7	15.6	0.245	-29.4	12.6
L3, scheduled	0.30	13.9	10.5	0.398	-0.9	7.4
L3, scheduled	0.40	10.5	11.4	0.368	-0.9	7.3

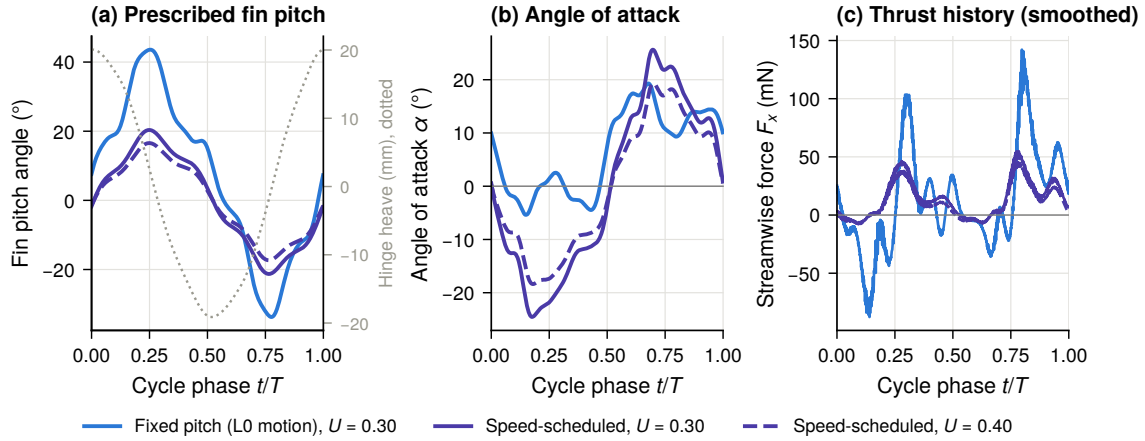


Figure 8.4: Fixed L0 motion versus speed-scheduled pitch. (a) Prescribed fluke pitch (hinge heave dotted, right axis). (b) Resulting angle of attack. (c) Streamwise force, smoothed.

range of 0.2–0.4 of efficient cruising in animals and close to the mean value measured for odontocetes [TNT03; RF04]; based on the trailing-edge excursion it is somewhat higher. The schedule needs no flow information, only $\dot{h}_{\max}$ and U . On the hardware it corresponds to a pitch amplitude that decreases with speed; how closely a passive spring can provide it is discussed in Section 9.3.

8.5 Self-Propulsion Estimate

Resistance. Three resistance curves bound the resistance of the robot without its propulsors (Figure 8.5a):

- $D_{\text{low}} = 10.8 \text{ mN} (U/0.2 \text{ m s}^{-1})^{1.72}$: the bare hull, from separate two-phase CFD of the hull at 0.1 m s^{-1} and 0.2 m s^{-1} ;
- $D_{\text{mid}} = D_{\text{low}} + 15.2 \text{ mN} (U/0.2 \text{ m s}^{-1})^2$: hull plus the submerged linkages, about 26 mN at 0.2 m s^{-1} ; the linkage term, equivalent to $C_D A \approx 7.6 \text{ cm}^2$, is an engineering estimate, not a measurement; this is the best estimate;
- $D_{\text{up}} = 77.6 \text{ mN} (U/0.2 \text{ m s}^{-1})^2$: the whole robot at rest with its fan paddles open, from CFD; it counts the paddles' own drag, which is already contained in $\bar{T}$, and is therefore an upper bound.

The thrust curves are interpolated with monotone cubic splines and extrapolated linearly beyond the last CFD point.

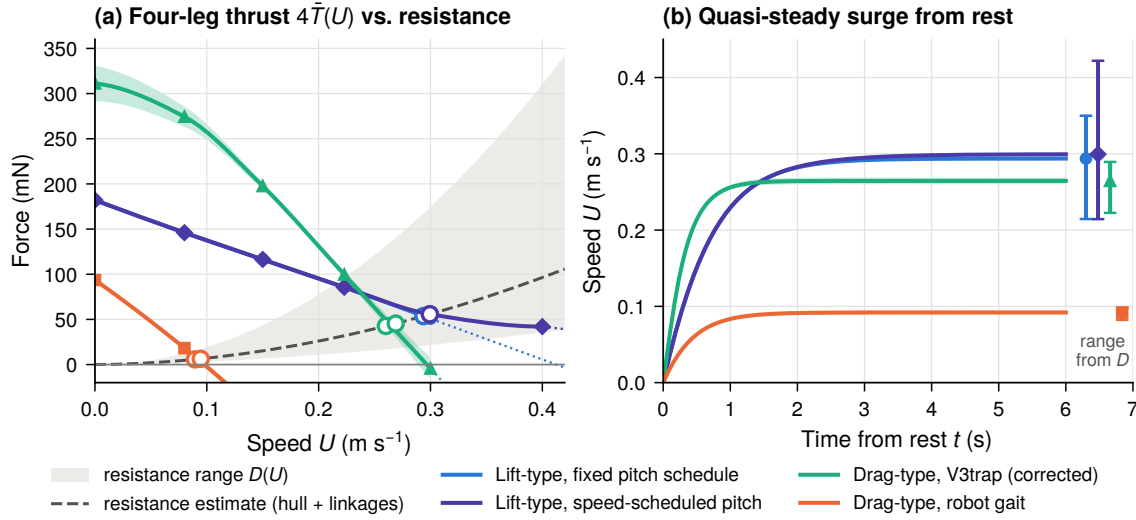


Figure 8.5: Self-propulsion estimate. (a) Four-leg thrust versus the resistance range; circles mark the cruising speeds for D_{mid} ; shaded: range of the added-mass correction of the paddle. Dotted: linear extrapolation. Below 0.30 m s^{-1} , where the speed-scheduled pitch was not simulated, its curve uses the fixed-pitch data. (b) Quasi-steady acceleration from rest for D_{mid} ; bars on the right give the range of cruising speeds from D_{up} to D_{low} .

Table 8.5: Predicted cruising speed and four-leg input power for D_{mid} , with the range from D_{up} to D_{low} . Paddle values are given as the range of the added-mass correction. * extrapolated beyond the CFD data. t_{90} : time to reach 90 % of the cruising speed from rest.

Propulsor	U_c (m s^{-1})	Range (m s^{-1})	$4\bar{P}$ (mW)	$4\bar{P}/U_c$ (J m^{-1})	t_{90} (s)
Drag, robot gait	0.089–0.095	0.082–0.098	23.3	0.25–0.26	0.94–0.99
Drag, V3trap	0.260–0.269	0.223–0.289	73–74	0.27–0.28	0.67–0.69
Lift, fixed pitch	0.294	0.214–0.350*	62.7	0.21	1.46
Lift, scheduled pitch	0.299	0.214–0.422*	42.0	0.14	1.57

Cruising speed. For the best-estimate resistance, the fluke propels the robot at 0.30 m s^{-1} , or 1.3 hull lengths per second, and the optimized paddle at $0.26\text{--}0.27 \text{ m s}^{-1}$ (Table 8.5). Both operating points lie just above the crossover, where the fluke is stronger, but close to it, so the difference in speed is small. The power differs clearly: the speed-scheduled fluke needs 42 mW, 33 % less than the fixed-pitch fluke (63 mW) and 43 % less than the paddle (73 mW), which is also slower. The robot’s current gait reaches only about 0.09 m s^{-1} . The ranges are wide because the resistance is uncertain, and which propulsor is faster depends on where the resistance curve crosses the thrust curves. With the high resistance D_{up} , the operating point moves below the crossover and the paddle is slightly faster ($0.22\text{--}0.23$ versus 0.21 m s^{-1}); with the low resistance D_{low} , it moves further above the crossover and the fluke is clearly faster ($0.35\text{--}0.42 \text{ m s}^{-1}$, extrapolated, versus $0.28\text{--}0.29 \text{ m s}^{-1}$).

Acceleration. The acceleration from rest is integrated with the measured mass of 0.419 kg plus 10 % added mass; the mass affects only the time scale, not the cruising speed. The paddle accelerates the robot faster: it reaches 90 % of its cruising speed after 0.68 s, the fluke after 1.46–1.57 s (Figure 8.5b). This reflects the paddle’s higher thrust at low speed and matches the observation that rowing appendages favour acceleration and flapping ones cruising [WW00].

8.6 First Hardware Test

The robot with flukes (Section 4.4) was timed in a 1 m pool: it started from rest with its stern against one wall and the time was stopped when its bow hit the opposite wall, a distance of 0.738 m (pool length minus the overall length of the hull, 262 mm). The stroke frequency was set to 2 and 4 Hz in the firmware; there was one hand-timed run for each setting. The same run was simulated with the quasi-steady model of Figure 8.5b, the L0 thrust at 2 Hz and the measured mass (Table 8.6).

Table 8.6: First hardware test: time and mean speed over 0.738 m from rest. Frequencies are the values set in the firmware. Model: L0 thrust at 2 Hz with the three resistance curves of Section 8.5.

	Frequency set (Hz)	Time (s)	Mean speed (m s^{-1})
Hardware	2	5.2	0.142
Hardware	4	2.8	0.264
Model, D_{low} / D_{mid} / D_{up}	2	2.9 / 3.2 / 3.9	0.25 / 0.23 / 0.19

Thrust. At 2 Hz, where the stroke is close to L0, the hardware needs 1.3 to 1.6 times as long as predicted with D_{mid} to D_{up} (1.8 times with D_{low} , which does not apply to the hardware). The resistance alone cannot explain this: with the CFD thrust, a resistance of 205 mN at 0.2 m s^{-1} would be required, 2.6 times D_{up} . This corresponds to $C_D A \approx 103 \text{ cm}^2$, or $C_D \approx 4$ on the submerged frontal area of the hull of 24 cm^2 , far above any bluff body. D_{up} itself corresponds to $C_D \approx 1.6$ and is a generous bound even for the hull moving blunt end first, while D_{low} , the bare hull moving pointed end first, does not apply to the hardware. Between D_{mid} and D_{up} , the run is reproduced with 27–47 % of the CFD thrust, roughly a quarter to a half, and a steady speed of about $0.16\text{--}0.20 \text{ m s}^{-1}$ at 2 Hz. Part of the deficit is expected from the stroke itself: the piecewise-linear trajectory has a peak heave speed about 15 % below that of L0 (Section 4.4), which costs of the order of 10–20 % of the thrust. The remaining factor of about two has several possible causes that the test cannot separate: the passive pitch, whose spring stiffness has not been verified against the angle of attack of L0; ventilation where the fluke breaks the surface; the reversed hull; and interaction between the legs. The self-propulsion speeds of Table 8.5 are therefore upper bounds for the present hardware.

Frequency. The ratio of the two runs does not depend on the thrust level. If the resistance grows with U^2 and the thrust scales with the square of the stroke velocity at a fixed Strouhal number, the time needed to cover a given distance from rest is inversely proportional to the product of frequency and amplitude. The measured ratio of 5.2 to 2.8 s, 1.86, is close to the value of 2 expected if the servos reach the full amplitude at 4 Hz, although the peak servo speed of about 800° s^{-1} , estimated from the firmware trajectory, is close to their no-load speed. A more precise statement is not possible: a timing error of 0.2 s per run moves the ratio between 1.67 and 2.08, and the passive pitch changes with frequency, because the hydrodynamic moment grows with the square of the stroke velocity while the spring moment does not. Taken at face value, the same scaling puts the steady speed at 4 Hz at roughly $0.3\text{--}0.4 \text{ m s}^{-1}$, above the CFD-based estimate at 2 Hz, for about six times the hydrodynamic power.

Chapter 9

Discussion

9.1 Evaluation of the Hypotheses

H1 (reduced-order optimization) is supported within the model. Whenever the model contains lift and it is not suppressed, all 101 feasible optima are lift-propelled. H1 is tested not only across models, where the lift-propelled gaits need one half to one third of the power of the best drag-only gaits, but also within one model: drag-based strokes forced by a lift-share limit need four to ten times the power and do not reach 0.30 m s^{-1} , although this second factor rests on only 3 feasible strokes. The statement is conditional on an uncalibrated lift model, as discussed below.

H2 (drag versus lift across speed) is supported. As expected, the optimized paddle is much stronger at low speed and its thrust falls steeply with speed, to about zero at 0.30 m s^{-1} ; the fluke keeps a larger fraction of its thrust and becomes stronger above $0.23\text{--}0.24 \text{ m s}^{-1}$. The cause of the paddle's speed limit is identified: the recovery drag, which grows more than sevenfold between rest and 0.30 m s^{-1} , and the momentum needed to open the fan in the oncoming flow, while the power-stroke thrust falls by less than half. The efficiency crossover lies slightly below the thrust crossover, at about 0.22 m s^{-1} ; below it, the optimized paddle is both stronger and more efficient than the fluke, reaching $\eta = 0.35\text{--}0.37$ at 0.15 m s^{-1} . This differs from the biological picture, in which lift-based swimmers are more efficient than paddlers at all speeds [Fis96]; for a small robot leg with a well-folding paddle it holds only above the crossover. The second part of H2 is confirmed: scheduling the fluke pitch so that the angle of attack stays near 20° raises the efficiency at 0.30 m s^{-1} from 0.24 to 0.40.

The paddle's efficiency exceeds the maximum of about 0.33 that Fish reported for paddling animals [Fis84; Fis96]. The two values are not directly comparable: the animal values are whole-animal estimates, while η here is the hydrodynamic Froude efficiency of an isolated propulsor whose folded fan produces only about one ninth of the force of the open fan ($a = 0.11$, against an area ratio of 0.445 for the muskrat). It also stays below the quasi-steady bound $U/v_p \approx 0.6$ at 0.15 m s^{-1} , with the mean power-stroke speed $v_p \approx 0.25 \text{ m s}^{-1}$ of the fan centroid.

H3 (whole robot) is largely supported. With the best-estimate resistance the fluke reaches 0.30 m s^{-1} and the paddle $0.26\text{--}0.27 \text{ m s}^{-1}$, so the speeds are similar but not equal, and the speed-scheduled fluke needs 43 % less power than the paddle (42 versus 73 mW). A first hardware test, however, is 1.3 to 1.6 times slower than predicted (Section 8.6). The predicted speeds are therefore upper bounds for the present robot, and the comparison of the propulsors rests on the CFD.

9.2 Why the Reduced-Order Model and the CFD Differ at Low Speed

The reduced-order model and the CFD agree that drag-based paddling has a speed limit and that lift-based flapping wins at cruising speed. They disagree at low speed: MuJoCo finds lift-propelled gaits more economical already at 0.10 m s^{-1} , by a factor of about two to four (up to ten at 0.20 m s^{-1}), whereas the CFD finds the optimized paddle more efficient below about 0.22 m s^{-1} . Four differences explain this.

1. *The lift model is optimistic.* The flat-plate law $c_l \sin 2\alpha$ produces lift without the induced drag and tip losses of a finite wing, and the flipper of the toy model has an aspect ratio of only 1.4. The validation case suggests that even at an aspect ratio of 6 the finite span is a likely cause of an efficiency deficit of up to 18 %, with tip losses estimated at 10–20 %. The model also has no dynamic stall, no delay in the build-up of circulation and no interaction of the flipper with its own wake.
2. *The drag model is pessimistic for a starting paddle.* A constant $C_D = 2.2$ misses the starting vortex and added-mass effects that raise the effective force coefficient of the paddle's power stroke to 2–3 times its quasi-steady value in the CFD (Section 8.1).
3. *The drag-type strokes differ.* The MuJoCo flipper can only feather by rotating; it has no folding and no controlled fold timing, and it is limited to the smooth harmonic motion (5.5). Because a flipper that crosses the water at an angle produces lift, a drag-based stroke in the lift model is a stroke from which lift has been suppressed, and most paddling strokes of the drag-only model become lift-propelled once lift is added (Section 5.4.3). The CFD paddle, in contrast, is a dedicated drag device: it moves broadside to its path, folds to 12 mm at optimized phases and uses a trapezoidal velocity law, each feature being worth up to about 30 % in thrust. MuJoCo thus compares lift-based strokes with lift-suppressed flipper strokes, the CFD a fluke with an optimized folding paddle.
4. *The optimization effort is asymmetric in both directions.* In MuJoCo the lift gaits were optimized for power at a required speed, while the drag-only front has not converged and the constrained search for drag-based strokes in the lift model found a feasible stroke in only 3 of 30 runs. In the CFD, the paddle was optimized in several rounds, while the fluke motion L0 was taken from another model; only its pitch amplitude was later scheduled with speed. The areas also differ (18.5 versus 11.2 cm^2).

The reduced-order model is therefore best read as an argument about *mechanism*, which propulsion principle an optimizer exploits when it can, and the CFD as the argument about *magnitude* and about the speed at which one principle overtakes the other. Where the two disagree, the CFD, which resolves the flow and is validated, is taken as authoritative. A calibration of the reduced-order coefficients, especially c_l with an aspect-ratio correction, against the CFD would reconcile the two and is the natural next step.

9.3 Implications for BODY2

Recommendation. For crossing open water at the surface, the recommended propulsor for BODY2 is the fluke with a pitch that decreases with speed. For the best-estimate resistance it is slightly faster than the optimized paddle V3trap, needs 43 % less hydrodynamic power, and requires neither a controlled fold nor a trapezoidal velocity law. Compared with the robot's present configuration, it roughly triples the predicted cruising speed, and in an untimed comparison on the hardware the flukes swam visibly faster than the paddles (Section 4.4). The paddle remains the better choice where acceleration, station keeping at low speed or walking matter; it then needs a fold narrowed to about 12 mm that can be commanded within 45–60 ms at a specific phase, which the current cable fold cannot do. This division of labour, rowing for starting and low speed, flapping for cruising, is the one found in animals [WW00; Fis96]; a leg that could change from paddle to fluke, like the morphing turtle limbs of Baines et al. [Bai+22], would cover both regimes.

Fluke. The fluke's thrust at rest is limited because it is stalled: without forward speed, a flapping foil acts as a small paddle. At speed, its performance depends on keeping the angle of attack near the efficient range, which a fixed pitch schedule does not do. The schedule $\theta_0 = \gamma_{\max}/2$ is a single-parameter rule that achieves this and also removes the large mean vertical force that a mismatched schedule produces. On the hardware the pitch is passive: it follows from the balance between the hydrodynamic moment about the pin, which turns the fluke towards the inflow, and the leaf spring. A passive fluke therefore adapts to speed, but not in the ratio required. The hydrodynamic moment grows with the square of the relative speed while the spring moment does not, so the ratio θ/γ rises with speed. A linear spring can match $\theta_0 = \gamma_{\max}/2$ at one design speed only; below it the fluke pitches too little and its angle of attack is too large, above it the fluke aligns too far with the inflow and its angle of attack falls below the efficient range. Matching a range of speeds requires a progressive spring, an adjustable preload or an active pitch actuator. The hardware test, which reached only roughly a quarter to a half of the CFD thrust (Section 8.6), is consistent with a passive pitch that is not tuned to the efficient angle of attack, although other causes cannot be excluded.

Equal area. The fluke used here has 61 % of the paddle's area. At rest both produce the same thrust per area (3.9–4.5 versus 4.1 mN cm⁻²), and at 0.223 m s⁻¹ the fluke already produces more (1.9 versus 1.3–1.4 mN cm⁻²). If the thrust and power coefficients stayed the same, a fluke of the paddle's area, 18.5 cm², moved with the same kinematics would produce about 75 mN at rest, as much as the paddle, and more than the paddle from about 0.15 m s⁻¹ on. Its power would grow in the same proportion, to about 30 mW at rest, so the efficiency crossover would not move. A lunate fluke of this area (span 86 mm, $AR \approx 4$) has been designed for the robot; its larger aspect ratio should reduce tip losses, but its larger hinge moment, about 1.6 times that of A2 at the same pitch schedule, must be carried by the pitch spring or actuator. This is an estimate, not a simulation result.

Actuators. The hydrodynamic moments of all motions are small compared with the servos: the largest, 36 mN m about the hip for the trapezoidal paddle stroke, is 6–8 % of the stall torque of the PTK 7465W (0.44–0.57 N m). The limits that matter are joint speed and the ability to switch the fold (Table A.3). The prescribed pitch of L0 needs 712 ° s⁻¹, more than the no-load speed at 6 V (650 ° s⁻¹) and close to that at 8.4 V (860 ° s⁻¹), whereas the speed-scheduled pitch at 0.30 m s⁻¹ needs only 366 ° s⁻¹.

9.4 Limitations

- *Single leg.* The CFD simulates one propulsor without the linkage, the hull or the other legs. Interaction between legs can raise the thrust of four paddling legs above four times the single-leg value [Wan+25a] and may differ between paddle and fluke. The self-propulsion balance neglects it.
- *Free surface.* All propulsor CFD is single-phase with a rigid lid. On the hardware and in Flare Live the fluke breaks the surface by about 7 mm at the top of its stroke, and the paddle's recovery path is close to it; ventilation and waves are not modelled.
- *Prescribed pitch and fold.* The fluke pitch is prescribed rigidly, while the hardware fluke is passive. The fan's folding is modelled as an instantaneous switch between two separately simulated states. The change of added mass at the switches is corrected analytically within bounds, with an added mass identified from the CFD; the effect of the open run's own recovery wake on its power stroke is not corrected and makes the paddle results slightly optimistic.
- *CFD accuracy.* The flow is laminar, which is one of the causes of the 18 % efficiency deficit in the validation at $Re = 4 \times 10^4$; at the lower Reynolds numbers of the fluke the effect is expected to be smaller. The fluke thrust carries about ± 12 % discretization uncertainty. The paddle mesh is coarser and was not studied separately; the conclusions were argued in Section 7.4 to be insensitive to an error of that size.
- *Power.* The power is the hydrodynamic input power of the propulsor. Servo losses, the inertia of the leg and the drag of the linkage are not included, so the power ratios between propulsors are not the ratios of battery power.
- *Unequal propulsors.* The paddle stroke was optimized more thoroughly than the fluke motion, so the fluke results are conservative; a systematic optimization of heave amplitude, frequency and pitch would likely move the crossover to lower speed. The propulsors also run at different frequencies (1.39 and 2 Hz) and have different areas; thrust per area and per power are reported, but no comparison at equal area and frequency was made.
- *Resistance.* The hull resistance is known only within bounds, and the cruising speeds are correspondingly uncertain. The linkage term of D_{mid} is an engineering estimate, not a measurement, and the hull resistance was computed for the pointed end first, whereas the hardware swims with its blunt end first.
- *Reduced-order model.* The MuJoCo model is a stand-in of similar size, not BODY2, and its coefficients are not calibrated. Its servo model (3 N m , 400° s^{-1}) is stronger and slower than the PTK 7465W ($0.44 - 0.57 \text{ N m}$, $650 - 860^\circ \text{ s}^{-1}$). Its searches are not fully converged, and the within-model comparison rests on 3 feasible drag-based strokes from 30 searches with two seeds, so its power factor may be overestimated.
- *Hardware.* The hardware test consists of one hand-timed run per frequency from rest, with a reversed hull, a passive pitch and an unmeasured stroke amplitude. It shows that the self-propulsion estimate is an upper bound but does not identify the causes of the difference.

Chapter 10

Conclusion and Outlook

10.1 Conclusion

This thesis compared drag-based paddling and lift-based flapping for the legs of the amphibious quadruped BODY2, using a reduced-order gait optimization and single-leg CFD of the real leg. Table 10.1 answers the three research questions.

Table 10.1: Answers to the research questions. Paddle: optimized fan paddle V3trap; fluke: A2 with the speed-scheduled pitch L3 unless stated otherwise; powers are hydrodynamic.

Question	Answer	Key numbers
RQ1: mechanism selected by the optimizer (Chapter 5)	Lift, whenever the model contains it	Unless lift is suppressed, all 101 feasible optima with lift are lift-propelled. Drag-based strokes need 2–10 times the power at equal speed: 2–3 times across models, 4–10 times within the lift model (3 feasible strokes of 30)
RQ2: drag versus lift across speed (Sections 8.1 to 8.4)	Paddle at low speed, fluke at cruising speed	At rest 73–83 versus 45.5 mN per leg, equal thrust per area. Crossover of thrust at $0.24 \pm 0.02 \text{ m s}^{-1}$, of efficiency at $0.22 \pm 0.02 \text{ m s}^{-1}$. Paddle thrust vanishes near 0.30 m s^{-1} . Scheduled pitch raises η at 0.30 m s^{-1} from 0.24 to 0.40
RQ3: whole robot (Sections 8.5 and 8.6)	Fluke slightly faster, much more economical; paddle accelerates faster	0.30 versus $0.26 - 0.27 \text{ m s}^{-1}$ at 42 versus 73 mW (–43 %); current gait about 0.09 m s^{-1} ; t_{90} 0.68 versus 1.46–1.57 s. Hardware at 2 Hz 1.3–1.6 times slower than predicted: the estimate is an upper bound

The CFD reproduces the thrust coefficient of a flapping-foil experiment to within 8 %, with about $\pm 12 \%$ discretization uncertainty in thrust. The paddle values are given as a range because the two-state model of the folding fan does not resolve the change of added mass at the fold, which is bounded analytically. The reduced-order model and the CFD agree on the mechanism and on the speed limit of paddling, but not on the speed below which the

paddle is more efficient; there the validated CFD is taken as authoritative. A first test of the robot with flukes reached roughly a quarter to a half of the CFD thrust, for reasons that the test cannot separate (Section 8.6), so the predicted cruising speeds are upper bounds for the present hardware.

Recommendation. For BODY2 swimming at the surface, the fluke with a pitch amplitude that decreases with speed is the better propulsor for cruising. The folding paddle is better for starting and low-speed manoeuvring, but only with a fold that narrows to about 12 mm and can be switched at a specific phase.

10.2 Outlook

- *Passive pitch.* Simulate the fluke with its real pin and leaf spring, coupling the pitch dynamics to the CFD forces, and design a progressive spring or preload that approximates the speed-scheduled pitch over the whole speed range.
- *Fairer fluke.* Optimize the fluke's heave amplitude, frequency and pitch phase with the same care as the paddle stroke, including a pitch near 45° at rest, and simulate the 18.5 cm^2 lunate fluke of the paddle's area (Section 9.3). Both are expected to move the crossover to lower speed.
- *Folding paddle.* Simulate the fold itself with a deforming or multi-body mesh to replace the two-state composite and its added-mass bounds.
- *Whole robot and free surface.* Simulate four legs with the hull, including the free surface, to quantify leg–leg interaction and surface effects on both propulsors.
- *Calibration.* Calibrate the reduced-order coefficients, in particular a lift slope with aspect-ratio correction, and the servo model against the single-leg CFD and the PTK 7465W, and repeat the gait optimization; give the drag-based strokes of Section 5.4.3 a fairer search.
- *Experiments.* Measure the resistance of the reversed hull in a coast-down test, the static thrust of the robot and its steady swimming speed from a flying start in repeated runs, and film the fluke pitch to find the cause of the thrust deficit.

Appendix A

Additional Data

A.1 Drag-Type Paddle: Power Stroke and Recovery

Table A.1: Contributions of power stroke and recovery to the mean thrust of the fan paddle, (3.12), in mN, for $\tau_o = 0$ and $\tau_c = 0.33$ (V3trap) or $\tau_c = \text{DUTY}$ (robot gait). Net: two-state composite; corrected: range of the added-mass correction (6.2).

$U \text{ (m s}^{-1}\text{)}$	power	V3trap				Robot gait		
		recovery	net	corrected		power	recovery	net
0	88.0	−5.3	82.7	72.9–82.7		29.4	−6.0	23.4
0.08	85.7	−10.2	75.6	65.8–71.4		18.1	−12.9	5.2
0.15	75.4	−17.2	58.2	48.4–50.4		12.6	−25.7	−13.1
0.223	62.5	−26.9	35.7	23.9–25.9		–	–	–
0.30	50.9	−39.1	11.7	−4.1–1.9		–	–	–

A.2 Fluke Cases

Table A.2: All single-leg fluke cases. α_{75} : 75th percentile of $|\alpha|$ over the cycle.

Case	Motion	$U \text{ (m s}^{-1}\text{)}$	$\bar{T} \text{ (mN)}$	$\bar{P} \text{ (mW)}$	η	α_{75}
L2U000	L0	0	45.5	18.1	–	73°
L2U008	L0	0.08	36.5	17.1	0.171	–
L2U015	L0	0.15	29.1	16.6	0.263	–
L0	L0	0.223	21.4	16.3	0.293	19.9°
L2U030	L0	0.30	12.7	15.6	0.245	13.7°
L1a10	α law 10°	0.223	≈ 31	≈ 35	≈ 0.20	13°
L1a28	α law 28°	0.223	26.2	23.9	0.244	30°
L1a20	α law 20°, paddle-leg hinge path	0.40	15.4	24.3	0.25	21.7°
L3U030	scheduled, $\theta_0 = 21.6^\circ$	0.30	13.9	10.5	0.398	21.0°
L3U040	scheduled, $\theta_0 = 17.6^\circ$	0.40	10.5	11.4	0.368	16.4°
L0c	L0, coarse mesh	0.223	−7.3% thrust in the common window (Section 7.2)			

A.3 Actuator Requirements

Table A.3: Peak joint speeds and peak hydrodynamic moments of the motions (propulsor only; inertia and drag of the linkage not included), compared with the servos of the robot.

	Paddle, V3trap	Fluke, L0	Fluke, L3 at 0.30 m s^{-1}
Leg joints	hip 214° s^{-1} , curl 459° s^{-1}	knee $\approx 310^\circ \text{ s}^{-1}$	as L0
Propulsor joint	fold within 45–60 ms at τ_c	pitch 712° s^{-1}	pitch 366° s^{-1}
Hydrodynamic moment	35.6 mN m about the hip	12.6 mN m about the pin (at 0.30 m s^{-1})	7.4 mN m about the pin
Available	PTK 7465W at 6.0–8.4 V: stall torque 0.44–0.57 N m, no-load speed 650–860 $^\circ \text{ s}^{-1}$; no controllable fold on the hardware		

Appendix B

Inter-Limb Coordination in the Reduced-Order Model

This appendix reports the results of the reduced-order model on the coordination of the four legs. They do not affect the comparison of drag- and lift-based propulsion in Chapter 5, but they show how strongly the attitude limits shape the optimal gait.

B.1 Gait Families without Lift

With drag only and moderate limits, all five families find feasible gaits (Table B.1, Figure B.1a). Bound is the fastest, $0.481 \pm 0.034 \text{ m s}^{-1}$ or about two body lengths per second, with a margin larger than the seed scatter. It is left–right symmetric, so heading and roll vanish and all degrees of freedom serve propulsion; all three bound solutions sit at the 15° pitch limit and at the upper frequency bound of 2.5 Hz. About two thirds of all solutions end within 1° of a limit, and 14 of 15 keep their servos at the torque limit throughout. The limits shape the answer and are reported with it.

With strict limits and drag only, the advantage of bound disappears: it drops to $0.159 \pm 0.020 \text{ m s}^{-1}$, all ten family solutions sit at the 5° pitch limit, and walk, pace and trot differ by less than the seed scatter (Figure B.1b). Pitch is the bottleneck of every coordination.

B.2 Coordination with Lift under Strict Limits

With lift and strict limits, all fifteen trot, pace and bound solutions over five seeds are lift-propelled, feasible and do not surface (Table B.1, Figure B.1c). Bound is again the fastest, up to 0.483 m s^{-1} for one seed, as fast as the fastest drag-only gait under moderate limits, but its servos are saturated almost all the time. Trot has the lowest mean power ($1.23 \pm 0.90 \text{ W}$) but is also the slowest of the three, so its low energy per metre mixes “economical” with “slow”.

To separate the two, the least power at equal speed was computed per family with lift and strict limits (five seeds, Figure B.3). At 0.10 m s^{-1} trot needs 0.10 W against 0.17 W for bound, and at 0.15 m s^{-1} 0.48 against 0.51 W; the medians over seeds are close. At 0.20 m s^{-1} trot needs 1.68 W, while pronk and bound need 0.96 and 0.98 W, and only three of five trot seeds

Table B.1: Fastest gait per family (speed in m s^{-1} , mean $\pm$ standard deviation over seeds). Moderate limits 10/10/15°, strict limits 10/5/5°. Seeds per family: 3 (drag only, moderate), 2 (drag only, strict), 5 for bound, pace and trot and 2 for walk and pronk (drag + lift, strict).

Family	Drag only, moderate	Drag only, strict	Drag + lift, strict
bound	0.481 ± 0.034	0.159 ± 0.020	0.295 ± 0.139
pace	0.345 ± 0.064	0.191 ± 0.062	0.204 ± 0.085
trot	0.344 ± 0.048	0.172 ± 0.013	0.139 ± 0.051
walk	0.344 ± 0.098	0.209 ± 0.089	0.143 ± 0.077
pronk	0.265 ± 0.038	0.145 ± 0.043	0.118 ± 0.064
free phase, 10^4 evaluations	–	–	0.303 ± 0.078

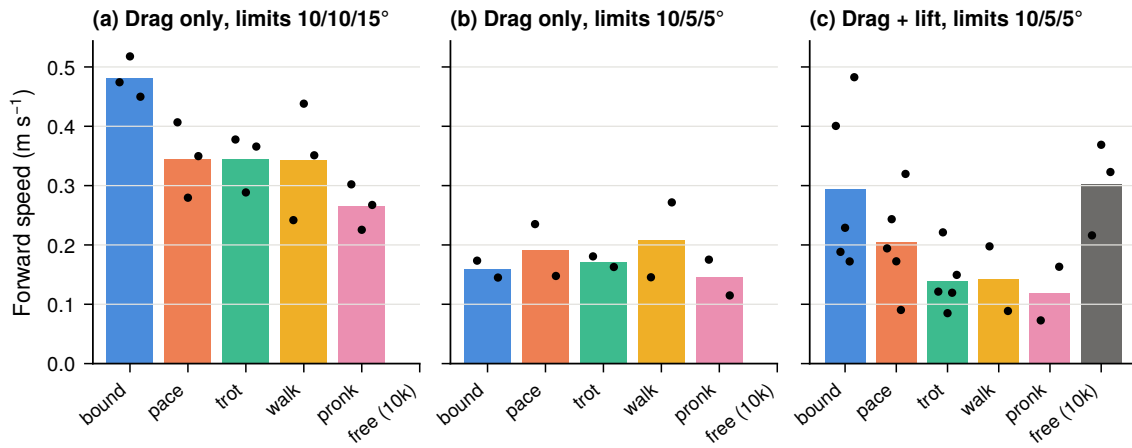


Figure B.1: Fastest gait per family in the reduced-order model; bars are seed means, dots individual seeds. (a) Drag only, moderate limits. (b) Drag only, strict limits. (c) Drag and lift, strict limits, including three free-phase searches with 10^4 evaluations. Free-phase searches with 2500 evaluations are not shown because they had not converged.

are feasible; trot's speed limit under strict limits is about 0.22 m s^{-1} . All 55 feasible solutions of this experiment are lift-propelled. Diagonal coordination is therefore at most marginally more economical at low speed; above about 0.15 m s^{-1} , fore-aft coordinations are both more economical and faster. Figure B.2 shows one lift-propelled trot at 0.221 m s^{-1} and 1.03 W , whose servos never saturate.

The free-phase search needs a larger budget. With 2500 evaluations it reached only $0.05 - 0.07 \text{ m s}^{-1}$; with 10^4 evaluations, three seeds reached $0.303 \pm 0.078 \text{ m s}^{-1}$, all lift-propelled and all converged to fore-aft coordinations close to bound or pronk (deviation 15–19 % of a period), not to trot. The best free gait, 0.369 m s^{-1} at 3.3 W , is faster than three of the five bound optima and needs less energy per metre than all five, which shows that better coordinations exist outside the fixed families.

B.3 Electrical Power

Because most of the fastest gaits keep their servos at the torque limit, the mechanical-power ranking could hide large copper losses. The electrical power was therefore recomputed for all 222 valid optima. Averaged per family, it lies within about $\pm 30\%$ of the mechanical power, because at saturation the back-EMF term cancels much of the stall torque. The exceptions are

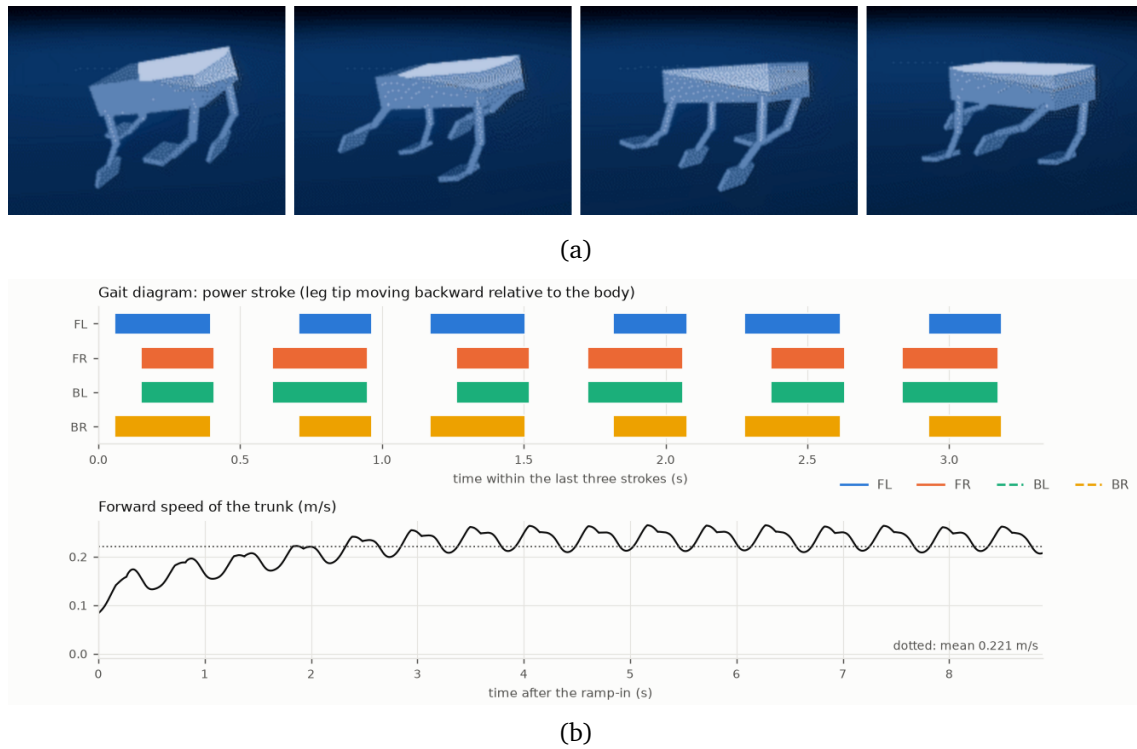


Figure B.2: A lift-propelled trot found with strict limits (0.90 Hz, DUTY = 0.59, 0.221 m s^{-1} , 1.03 W). (a) Four frames of the MuJoCo animation, 0.25–0.30 s apart, covering one stroke period of 1.11 s; the camera follows the trunk. (b) Top: power strokes of the four legs over the last three cycles; diagonal legs act together. Bottom: forward speed of the trunk after the ramp-in, mean dotted.

trot and pronk with drag only under moderate limits, whose electrical power is 71 % and 44 % higher (for trot 19.6 instead of 11.5 W). This moves trot behind pace and walk in energy per metre; apart from this, the statements of this appendix hold for the electrical power as well.

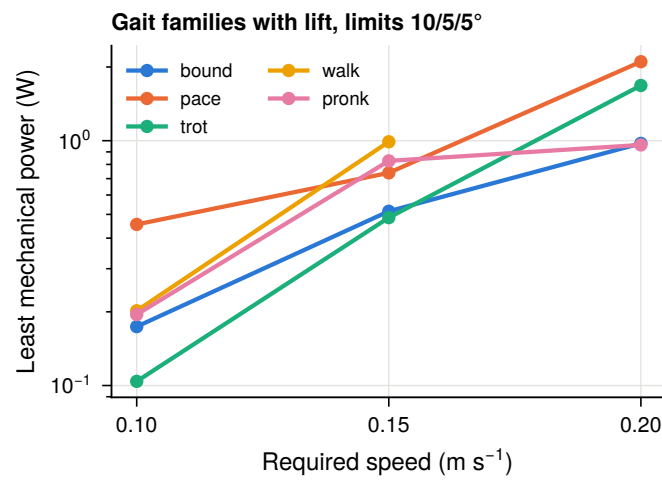


Figure B.3: Least mechanical power to reach a required speed per gait family, with lift and strict limits (best of five seeds).

Appendix C

Real-Time Solver: Tow Runs and Whole-Robot Trends

This appendix collects the Flare Live results. The tow runs are the basis of the cross-check in Section 7.5; the self-propelled runs show trends of the whole robot but no reliable magnitudes.

C.1 Tow Runs

Table C.1: Flare Live tow runs with the L0 motion: streamwise fluid force on the fluke (fluke frame) and net thrust of the whole flapping leg, in mN, averaged over $t = 3 - 9$ s.

U (m s ⁻¹)	Lattice	Fluke, flapping	Fluke, idle	Leg net thrust	CFD fluke	α_{75}
0	3 mm	20.0	–	25.7	45.5	73.8°
0.1	3 mm	8.5	–	15.9	≈ 34	41.1°
0.223	3 mm	–0.1	–12.6	9.6	21.4	19.7°
0.223	2 mm	4.1	–11.6	5.8	21.4	19.8°

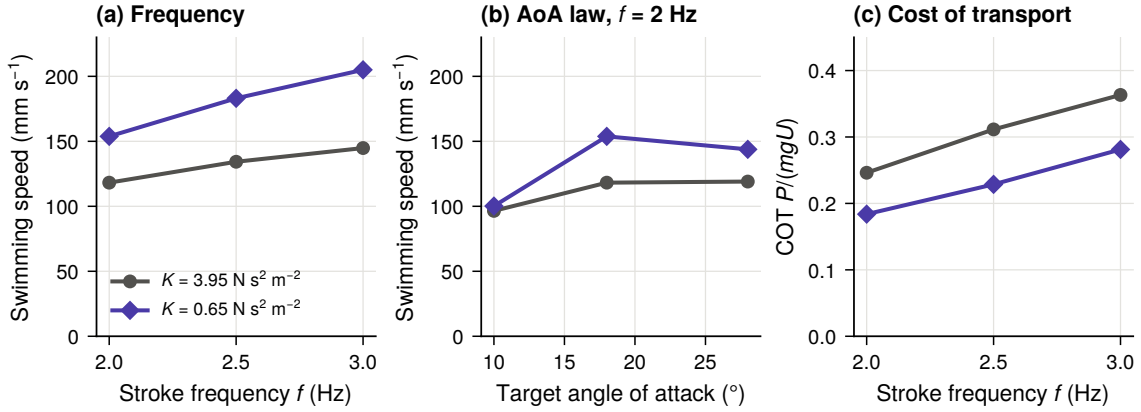
C.2 Self-Propelled Runs

Because of the thrust deficit of Section 7.5, the self-propelled Flare Live runs are used only for trends (Figure C.1, Table C.2). With the resistance $K = 0.65 \text{ N s}^2 \text{ m}^{-2}$ (D_{mid}), a trot at 2 Hz with a target angle of attack of 18° swims at 0.154 m s^{-1} , about half the CFD-based prediction, consistent with the missing fluke thrust.

- *Frequency.* Speed grows with frequency as $U \propto f^{0.71}$ at $K = 0.65$ ($f^{0.51}$ at $K = 3.95$), while power grows as $f^{1.8}$; the cost of transport rises with frequency. The Strouhal number stays nearly constant.
- *Angle of attack.* The target 18° gives the highest speed and the lowest cost of transport; 28° is slightly slower, 10° much slower. The controller does hold the target: in the 10° run, the mid-stroke angle of attack has a median of 10.1°, but the fluke has to rotate by 122° per cycle and sits at its end stop 11 % of the time. In CFD, the 10° law gave

Table C.2: Self-propelled Flare Live runs, trot, $K = 0.65 \text{ N s}^2 \text{ m}^{-2}$. $\text{COT} = P_j/(mgU)$ with the joint-drive power P_j and $m = 0.863 \text{ kg}$.

Run	U (mm s^{-1})	P_j (mW)	COT
2 Hz, $\alpha 10^\circ$	100.2	261	0.31
2 Hz, $\alpha 18^\circ$	153.8	239	0.18
2 Hz, $\alpha 28^\circ$	143.8	249	0.20
2.5 Hz, $\alpha 18^\circ$	183.1	355	0.23
3 Hz, $\alpha 18^\circ$	205.0	488	0.28

**Figure C.1:** Self-propelled robot in Flare Live (trot). (a) Speed versus stroke frequency for two hull resistances. (b) Speed versus the target angle of attack at 2 Hz. (c) Cost of transport versus frequency.

the most thrust at this speed (Section 8.2). The penalty for small angles in Flare Live is consistent with its unresolved lift; only the statement that efficiency is highest near $18-20^\circ$ is supported by both solvers.

- *Resistance.* Lowering K from 0.65 to 0.27 (bare hull) raises the speed only from 0.154 to 0.160 m s^{-1} , so in Flare Live the robot is limited by its thrust rather than by its hull.
- *Attitude.* The hull heaves by less than 0.6 mm and pitches by less than 0.3° in the trot: diagonal coordination keeps the floating robot level.

The joint-drive power in Flare Live ($240-490 \text{ mW}$) is an order of magnitude above the hydrodynamic power of the CFD (about 65 mW for four flukes), because it includes the power to move the linkages, their inertia and the controller effort.

Appendix D

Reproducibility

D.1 Software

- OpenFOAM v2606, overPimpleDyMFoam; meshes with blockMesh and snappyHexMesh. Case generators, run scripts and post-processing in Python (make_leg_cfd.py, make_fluke_cfd.py, fluke_traj.py, fluke_forces.py).
- MuJoCo 3.14 with the Python package cma; the gait-optimization code swimopt (<https://github.com/lxtuni/swimopt>), including the raw results of all 234 runs, 222 of them valid (thesis_notes/data/all_runs.csv). The replay of the drag-only optima with lift (Section 5.4.3) is documented in the notes of the repository, not in all_runs.csv.
- Flare Live with the scripts run_float.py (self-propelled runs) and bench_foil.py (tow runs).

[**TODO:** Add the repository of the CFD and Flare Live scripts and the commit hashes of the final versions.]

D.2 MuJoCo Settings

Table D.1: Settings of the reduced-order gait optimization.

Item	Value
Time step, integrator	2 ms, implicit-fast
Evaluation	settle 1.0 s, ramp-in 1.5 s, score 8 s rounded up to whole periods
Servo	position servo, stiffness 60, critical damping, $\tau_s = 3 \text{ N m}$, $\omega_{\max} = 400^\circ \text{ s}^{-1}$
Flipper coefficients	$C_D = (2.2, 0.10, 0.10)$, $C_a = 1.0$, $c_v = 0.05$, $c_l = 1.1$
Link coefficients	$C_D = (0.5, 0.5, 0.25)$, $C_a = 0.3$, $c_v = 0.02$
Trunk coefficients	$C_D = (0.9, 1.6, 1.8)$, $C_a = 0.2$, $c_v = 0.10$
Gait	$H = 2$; $f \in [0.2, 2.5] \text{ Hz}$; DUTY $\in [0.25, 0.75]$; $A \in [0, 0.9] \text{ rad}$; $q_0 \in [-0.5, 0.5] \text{ rad}$
CMA-ES	population 16, $\sigma_0 = 0.25$, 2500 evaluations (10^4 for free phase), non-zero seeds
Limits	heading/roll/pitch RMS 10/10/15° or 10/5/5°; surfacing $\leq 5\%$ where stated; lift thrust share $s_L \leq 0.1$ for the drag-based strokes of Section 5.4.3

D.3 Model Errors Found and Corrected

Several errors in the simulation set-up were found during the work and corrected before the reported results were produced. They are listed because they are easy to repeat.

- MJCF reads joint ranges in degrees by default; a range of ± 1.3 intended in radians locked every joint to $\pm 1.3^\circ$. All results before this correction were discarded.
- Changing body masses at run time without recomputing MuJoCo's derived constants made the added mass ineffective for a single free body and caused a 36 % error in a momentum-conservation test.
- Visual and collision geometries imported from a URDF were both given hydrodynamic forces, doubling buoyancy, drag and added mass.
- An additive weighting of speed and attitude produced strongly rolling and yawing optima once the legs could move freely; it was replaced by the feasibility-first formulation (5.7).
- A first measure of the lift share, $|I_L|/(|I_L| + |I_{\text{drag}}|)$, is about 0.5 for every gait with lift, because in steady swimming the lift and drag impulses cancel. Used as a limit, it could only be met by gaits that barely moved. It was replaced by s_L of (5.3), and the runs with the old measure were discarded.
- In the fluke CFD, a coarse background mesh that had not been refined around the fluke led to a failed hole cutting in which almost the whole background was removed; the number of hole cells is now checked automatically.

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